

33. Realtime Clock (RTCd)

In this section, “PCLK” is used to refer to PCLKB.

33.1 Overview

The RTC has two types of counting modes: calendar count mode and binary count mode. They are used by switching the register settings.

For calendar count mode, the RTC has a 100 year calendar from 2000 to 2099 and automatically adjusts dates for leap years.

For binary count mode, the RTC does not count in terms of years, months, dates, day-of-week, hours, or minutes; it counts seconds, and retains the information as a serial value. This mode can be used for calendars other than the Gregorian calendar.

The count source of the time counters is selectable as the sub-clock or main clock.

The RTC uses the 128-Hz clock which is acquired by the count source divided by the prescaler as the reference clock.

Year, month, date, day-of-week, a.m./p.m. (in 12-hour mode), hour, minute, second, or 32-bit binary is counted in 1/128 second units.

Table 33.1 lists the specifications of the RTC, Figure 33.1 shows a block diagram of the RTC, and Table 33.2 shows the pin configuration of the RTC.

Table 33.1 RTC Specifications

Item	Description
Count mode	Calendar count mode/binary count mode
Count source*1	Sub-clock (XCIN) or main clock (EXTAL)
Clock and calendar functions	• Calendar count mode Year, month, date, day-of-week, hour, minute, second are counted, BCD display 12 hours/24 hours mode switching function 30 seconds adjustment function (a number less than 30 is rounded down to 00 seconds, and 30 seconds or more are rounded up to one minute) Automatic adjustment function for leap years • Binary count mode Count seconds in 32 bits, binary display • Common to both modes Start/stop function The sub-second digit is displayed in binary units (1 Hz, 2 Hz, 4 Hz, 8 Hz, 16 Hz, 32 Hz, or 64 Hz). Clock error correction function Clock (1 Hz/64 Hz) output
Interrupts	• Alarm interrupt (ALM) As an alarm interrupt condition, selectable which of the below is compared with: - Calendar count mode: Year, month, date, day-of-week, hour, minute, or second can be selected - Binary count mode: Each bit of the 32-bit binary counter • Periodic interrupt (PRD) 2 seconds, 1 second, 1/2 second, 1/4 second, 1/8 second, 1/16 second, 1/32 second, 1/64 second, 1/128 second, or 1/256 second can be selected as an interrupt period. • Carry interrupt (CUP) An interrupt is generated at either of the following timings: - When a carry from the 64-Hz counter to the second counter is generated. - When the 64-Hz counter is changed and the R64CNT register is read at the same time. • Recovery from software standby mode or deep software standby mode can be performed by an alarm interrupt or periodic interrupt
Time capture function	• Times can be captured when the edge of the time capture event input pin is detected. For every event input, month, date, hour, minute, and second are captured or 32-bit binary counter value is captured.
Event link function	Periodic event output

Note 1. Satisfy the frequency of the peripheral module clock (PCLKB) $\geq$ the frequency of the count source.

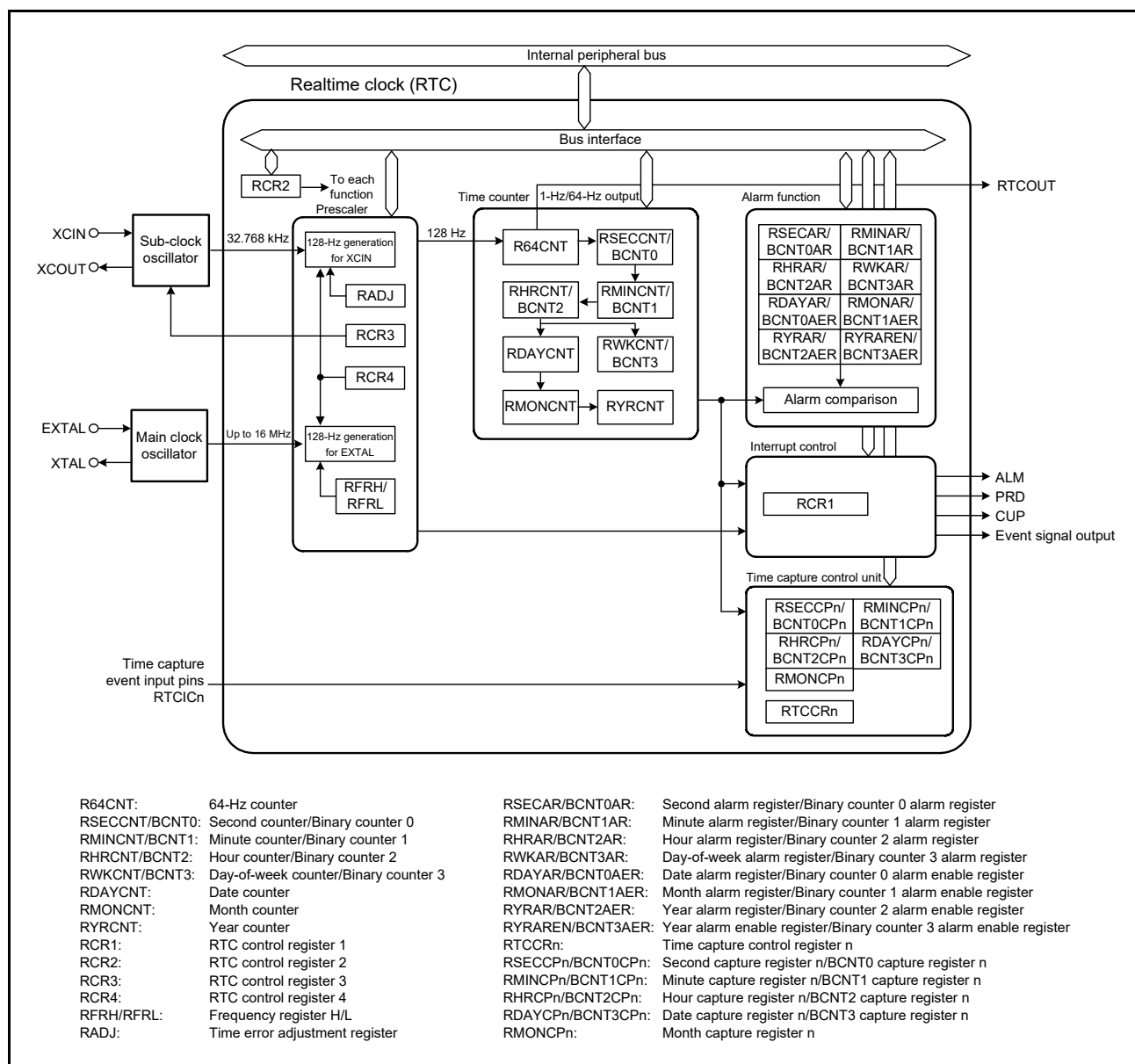


Figure 33.1 Block Diagram of RTC (n = 0 to 2)

Table 33.2 Pin Configuration of RTC

Pin Name	I/O	Function
XTAL	Output	These pins are used to connect a crystal.
EXTAL	Input	The EXTAL pin can also be used to input an external clock. For details, refer to section 9.3.2, External Clock Input.
XCIN	Input	Connect a 32.768-kHz crystal to these pins.
XCOUT	Output	
RTCOUT	Output	This pin is used to output a 1-Hz/64-Hz waveform, but not in deep software standby mode.
RTCIC0	Input	Time capture event input pins
RTCIC1	Input	
RTCIC2	Input	

33.2 Register Descriptions

When writing to or reading from RTC registers, do so in accordance with section 33.6.5, Notes on Writing to and Reading from Registers.

If the value in an RTC register after a reset is given as x (undefined bits) in the list, it is not initialized by a reset. When RTC enters the reset state or a low power consumption state during counting operations (i.e. while the RCR2.START bit is 1), the year, month, day of the week, date, hours, minutes, seconds, and 64-Hz counters continue to operate. However, when the count source is the main clock, the main clock is not stopped by a reset other than a deep software standby reset. For details, refer to Table 6.2, Targets to be Initialized by Each Reset Source. Note that a reset generated during writing to or updating of a register might destroy the register value. In addition, do not allow the chip to enter software standby mode or deep software standby mode immediately after setting any of these registers. For details, refer to section 33.6.4, Transitions to Low Power Consumption Modes after Setting Registers.

33.2.1 64-Hz Counter (R64CNT)

Address(es): RTC.R64CNT 0008 C400h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	F1HZ	F2HZ	F4HZ	F8HZ	F16HZ	F32HZ	F64HZ
Value after reset:	0	x	x	x	x	x	x	x
x: Undefined								

Bit	Symbol	Bit Name	Description	R/W
b0	F64HZ	64 Hz	Indicate the state between 1 Hz and 64 Hz of the sub-second digit.	R
b1	F32HZ	32 Hz		R
b2	F16HZ	16 Hz		R
b3	F8HZ	8 Hz		R
b4	F4HZ	4 Hz		R
b5	F2HZ	2 Hz		R
b6	F1HZ	1 Hz		R
b7	—	Reserved	This bit is read as 0. Writing to this bit has no effect.	R

The R64CNT counter is used in both calendar count mode and in binary count mode.

The 64-Hz counter (R64CNT) generates a period of one second by counting the 128-Hz reference clock.

The state in the sub-second range can be confirmed by reading this counter.

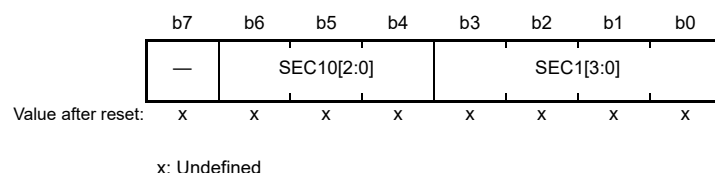
This counter is set to 00h by an RTC software reset or executing 30-second adjustment.

To read this counter, follow the procedure in section 33.3.5, Reading 64-Hz Counter and Time.

33.2.2 Second Counter (RSECCNT)/Binary Counter 0 (BCNT0)

(1) In calendar count mode:

Address(es): RTC.RSECCNT 0008 C402h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	SEC1[3:0]	1-Second Count	Counts from 0 to 9 every second. When a carry is generated, 1 is added to the tens place.	R/W
b6 to b4	SEC10[2:0]	10-Second Count	Counts from 0 to 5 for 60-second counting.	R/W
b7	—	Reserved	Set this bit to 0. It is read as the set value.	R/W

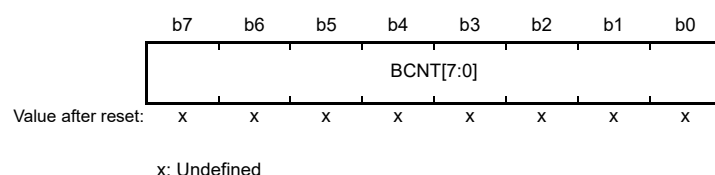
The RSECCNT counter is used for setting and counting the BCD-coded second value. It counts carries generated once per second in the 64-Hz counter.

The setting range is decimal 00 to 59. The RTC will not operate normally if any other value is set. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2.

After writing to the RSECCNT register, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

(2) In binary count mode:

Address(es): RTC.BCNT0 0008 C402h



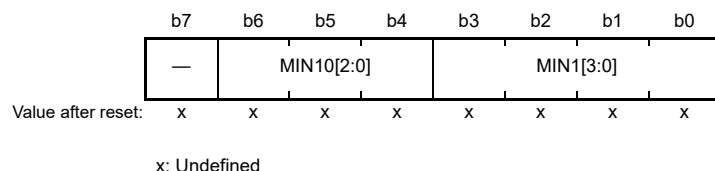
The BCNT0 counter is a readable/writable 32-bit binary counter b7 to b0.

The 32-bit binary counter performs count operation by a carry generating for each second of the 64-Hz counter. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2. To read this counter, follow the procedure in section 33.3.5, Reading 64-Hz Counter and Time.

33.2.3 Minute Counter (RMINCNT)/Binary Counter 1 (BCNT1)

(1) In calendar count mode:

Address(es): RTC.RMINCNT 0008 C404h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	MIN1[3:0]	1-Minute Count	Counts from 0 to 9 every minute. When a carry is generated, 1 is added to the tens place.	R/W
b6 to b4	MIN10[2:0]	10-Minute Count	Counts from 0 to 5 for 60-minute counting.	R/W
b7	—	Reserved	Set this bit to 0. It is read as the set value.	R/W

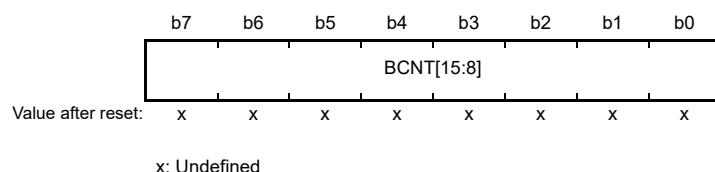
The RMINCNT counter is used for setting and counting the BCD-coded minute value. It counts carries generated once per minute in the second counter.

A value from 00 through 59 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2.

After writing to the RMINCNT counter, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

(2) In binary count mode:

Address(es): RTC.BCNT1 0008 C404h



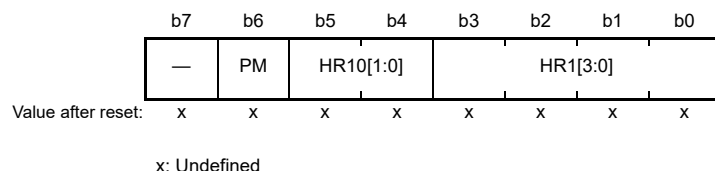
The BCNT1 counter is a readable/writable 32-bit binary counter b15 to b8.

The 32-bit binary counter performs count operation by a carry generating for each second of the 64-Hz counter. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2. To read this counter, follow the procedure in section 33.3.5, Reading 64-Hz Counter and Time.

33.2.4 Hour Counter (RHRCNT)/Binary Counter 2 (BCNT2)

(1) In calendar count mode:

Address(es): RTC.RHRCNT 0008 C406h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	HR1[3:0]	1-Hour Count	Counts from 0 to 9 once per hour. When a carry is generated, 1 is added to the tens place.	R/W
b5, b4	HR10[1:0]	10-Hour Count	Counts from 0 to 2 once per carry from the ones place.	R/W
b6	PM	PM	Time Counter Setting for a.m./p.m. 0: a.m. 1: p.m.	R/W
b7	—	Reserved	Set this bit to 0. It is read as the set value.	R/W

The RHRCNT counter is used for setting and counting the BCD-coded hour value. It counts carries generated once per hour in the minute counter.

The specifiable time differs according to the setting in the hours mode bit (RCR2.HR24).

When the RCR2.HR24 bit is 0: From 00 to 11 (in BCD)

When the RCR2.HR24 bit is 1: From 00 to 23 (in BCD)

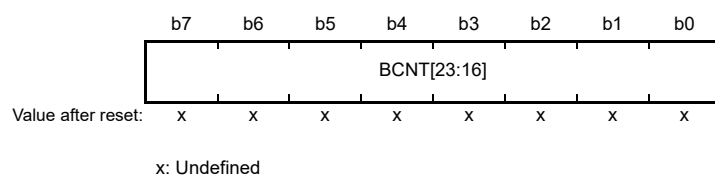
If a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2.

The PM bit is only enabled when the RCR2.HR24 bit is 0. Otherwise, the setting in the PM bit has no effect.

After writing to the RHRCNT counter, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

(2) In binary count mode:

Address(es): RTC.BCNT2 0008 C406h



The BCNT2 counter is a readable/writable 32-bit binary counter b23 to b16.

The 32-bit binary counter performs count operation by a carry generating for each second of the 64-Hz counter.

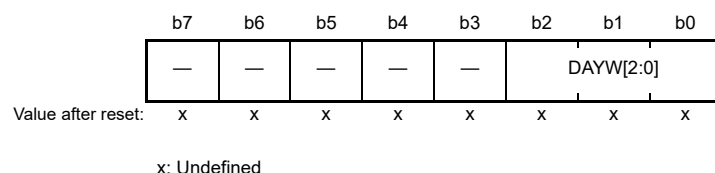
Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2.

To read this counter, follow the procedure in section 33.3.5, Reading 64-Hz Counter and Time.

33.2.5 Day-of-Week Counter (RWKCNT)/Binary Counter 3 (BCNT3)

(1) In calendar count mode:

Address(es): RTC.RWKCNT 0008 C408h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	DAYW[2:0]	Day-of-Week Counting	b2 b0 0 0 0: Sunday 0 0 1: Monday 0 1 0: Tuesday 0 1 1: Wednesday 1 0 0: Thursday 1 0 1: Friday 1 1 0: Saturday 1 1 1: Setting Prohibited	R/W
b7 to b3	—	Reserved	Set these bits to 0. They are read as the set value.	R/W

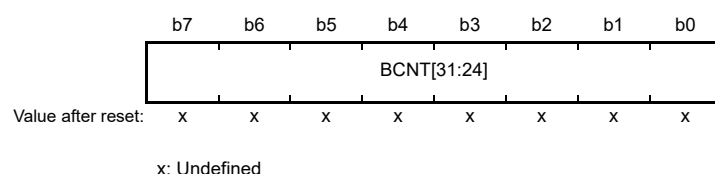
The RWKCNT counter is used for setting and counting in the coded day-of-week value. It counts carries generated once per day in the hour counter.

A value from 0 through 6 can be specified; if a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2.

Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

(2) In binary count mode:

Address(es): RTC.BCNT3 0008 C408h

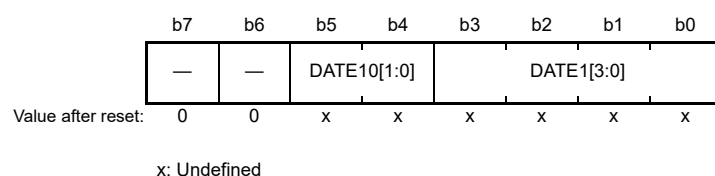


The BCNT3 counter is a readable/writable 32-bit binary counter b31 to b24.

The 32-bit binary counter performs count operation by a carry generating for each second of the 64-Hz counter. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2. To read this counter, follow the procedure in section 33.3.5, Reading 64-Hz Counter and Time.

33.2.6 Date Counter (RDAYCNT)

Address(es): RTC.RDAYCNT 0008 C40Ah



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	DATE1[3:0]	1-Day Count	Counts from 0 to 9 once per day. When a carry is generated, 1 is added to the tens place.	R/W
b5, b4	DATE10[1:0]	10-Day Count	Counts from 0 to 3 once per carry from the ones place.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The RDAYCNT counter is used in calendar count mode.

The RDAYCNT counter is used for setting and counting the BCD-coded date value. It counts carries generated once per day in the hour counter. The count operation depends on the month and whether the year is a leap year.

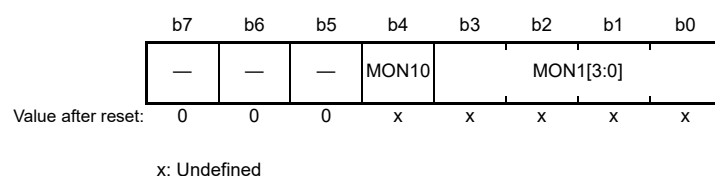
Leap years are determined according to whether the year counter (RYRCNT) value is divisible by 400, 100, and 4.

A value from 01 through 31 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly. (When specifying a value, note that the range of specifiable days depends on the month and whether the year is a leap year.) Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2.

After writing to the RHRCNT counter, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

33.2.7 Month Counter (RMONCNT)

Address(es): RTC.RMONCNT 0008 C40Ch



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	MON1[3:0]	1-Month Count	Counts from 0 to 9 once per month. When a carry is generated, 1 is added to the tens place.	R/W
b4	MON10	10-Month Count	Counts from 0 to 1 once per carry from the ones place.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The RMONCNT counter is used in calendar count mode.

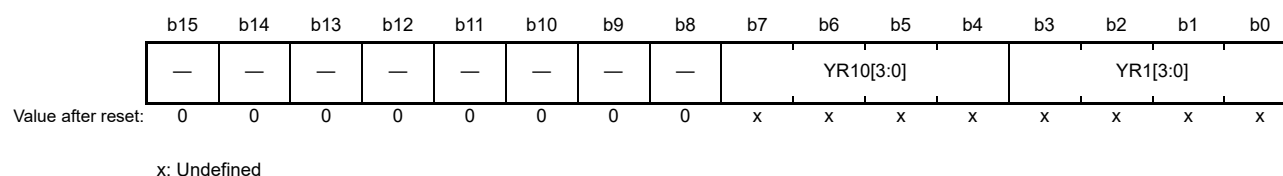
The RMONCNT counter is used for setting and counting the BCD-coded month value. It counts carries generated once per month in the date counter.

A value from 01 through 12 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2.

After writing to the RMONCNT counter, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

33.2.8 Year Counter (RYRCNT)

Address(es): RTC.RYRCNT 0008 C40Eh



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	YR1[3:0]	1-Year Count	Counts from 0 to 9 once per year. When a carry is generated, 1 is added to the tens place.	R/W
b7 to b4	YR10[3:0]	10-Year Count	Counts from 0 to 9 once per carry from ones place. When a carry is generated in the tens place, 1 is added to the hundreds place.	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The RYRCNT counter is used in calendar count mode.

The RYRCNT counter is used for setting and counting the BCD-coded year value. It counts carries generated once per year in the month counter.

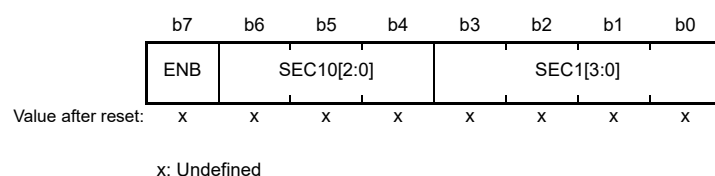
A value from 00 through 99 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2.

After writing to the RYRCNT counter, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

33.2.9 Second Alarm Register (RSECAR)/ Binary Counter 0 Alarm Register (BCNT0AR)

(1) In calendar count mode:

Address(es): RTC.RSECAR 0008 C410h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	SEC1[3:0]	1 Second	Value for the ones place of seconds	R/W
b6 to b4	SEC10[2:0]	10 Seconds	Value for the tens place of seconds	R/W
b7	ENB	ENB	0: The register value is not compared with the RSECCNT counter value. 1: The register value is compared with the RSECCNT counter value.	R/W

The RSECAR register is an alarm register corresponding to the BCD-coded second counter RSECCNT. When the ENB bit is set to 1, the RSECAR value is compared with the RSECCNT value. From among the alarm registers (RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, and RYRAREN), only those selected with the ENB bits set to 1 are compared with the corresponding counters. When the respective values all match, the IR flag corresponding to the ALM interrupt is set to 1.

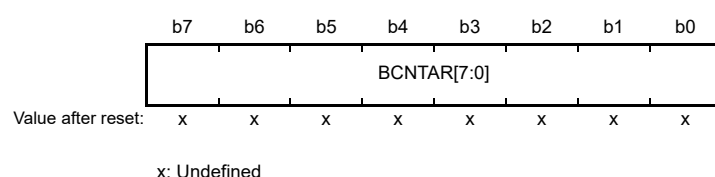
RSECAR values from 00 through 59 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly.

After writing to the RSECAR register, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

This register is set to 00h by an RTC software reset.

(2) In binary count mode:

Address(es): RTC.BCNT0AR 0008 C410h

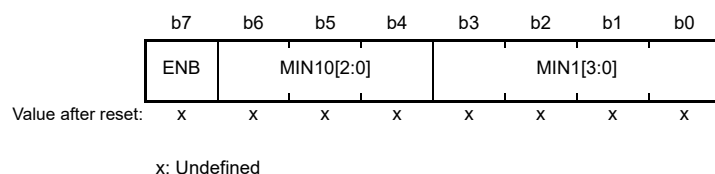


The BCNT0AR counter is a readable/writable alarm register corresponding to 32-bit binary counter b7 to b0. This register is set to 00h by an RTC software reset.

33.2.10 Minute Alarm Register (RMINAR)/ Binary Counter 1 Alarm Register (BCNT1AR)

(1) In calendar count mode:

Address(es): RTC.RMINAR 0008 C412h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	MIN1[3:0]	1 Minute	Value for the ones place of minutes	R/W
b6 to b4	MIN10[2:0]	10 Minutes	Value for the tens place of minutes	R/W
b7	ENB	ENB	0: The register value is not compared with the RMINCNT counter value. 1: The register value is compared with the RMINCNT counter value.	R/W

The RMINAR register is an alarm register corresponding to the BCD-coded minute counter RMINCNT. When the ENB bit is set to 1, the RMINAR value is compared with the RMINCNT value. From among the alarm registers (RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, and RYRAREN), only those selected with the ENB bits set to 1 are compared with the corresponding counters. When the respective values all match, the IR flag corresponding to the ALM interrupt is set to 1.

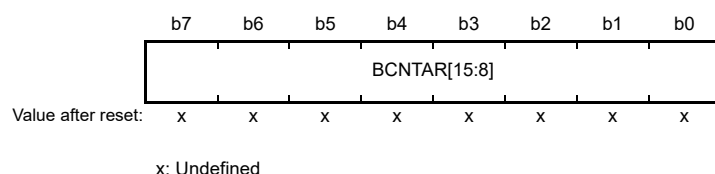
RMINAR values from 00 through 59 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly.

After writing to the RMINAR register, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

This register is set to 00h by an RTC software reset.

(2) In binary count mode:

Address(es): RTC.BCNT1AR 0008 C412h

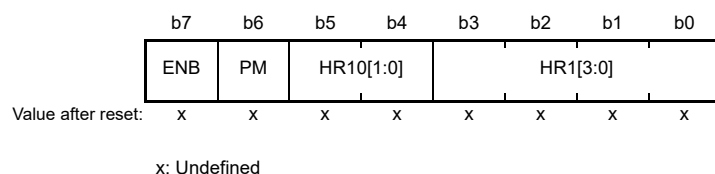


The BCNT1AR counter is a readable/writable alarm register corresponding to 32-bit binary counter b15 to b8. This register is set to 00h by an RTC software reset.

33.2.11 Hour Alarm Register (RHRAR)/ Binary Counter 2 Alarm Register (BCNT2AR)

(1) In calendar count mode:

Address(es): RTC.RHRAR 0008 C414h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	HR1[3:0]	1 Hour	Value for the ones place of hours	R/W
b5, b4	HR10[1:0]	10 Hours	Value for the tens place of hours	R/W
b6	PM	PM	Time Alarm Setting for a.m./p.m. 0: a.m. 1: p.m.	R/W
b7	ENB	ENB	0: The register value is not compared with the RHRCNT counter value. 1: The register value is compared with the RHRCNT counter value.	R/W

The RHRAR register is an alarm register corresponding to the BCD-coded hour counter RHRCNT. When the ENB bit is set to 1, the RHRAR value is compared with the RHRCNT value. From among the alarm registers (RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, and RYRAREN), only those selected with the ENB bits set to 1 are compared with the corresponding counters. When the respective values all match, the IR flag corresponding to the ALM interrupt is set to 1.

The specifiable time differs according to the setting in the hours mode bit (RCR2.HR24).

When the RCR2.HR24 bit is 0: From 00 to 11 (in BCD)

When the RCR2.HR24 bit is 1: From 00 to 23 (in BCD)

If a value outside of this range is specified, the RTC does not operate correctly.

When the RCR2.HR24 bit is 0, be sure to set the PM bit.

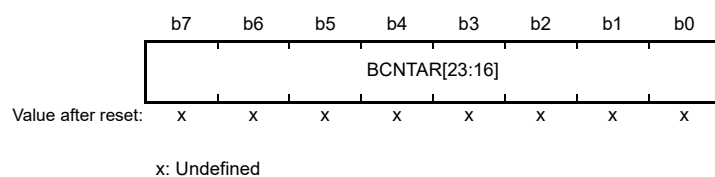
When the RCR2.HR24 bit is 1, the setting in the PM bit has no effect.

After writing to the RHRAR register, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

This register is set to 00h by an RTC software reset.

(2) In binary count mode:

Address(es): RTC.BCNT2AR 0008 C414h



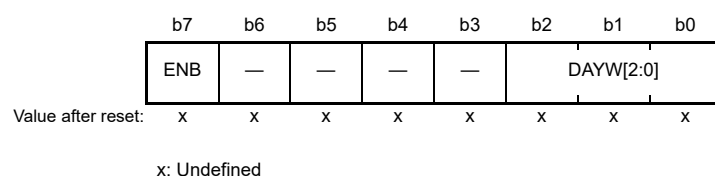
The BCNT2AR counter is a readable/writable alarm register corresponding to 32-bit binary counter b23 to b16.

This register is set to 00h by an RTC software reset.

33.2.12 Day-of-Week Alarm Register (RWKAR)/ Binary Counter 3 Alarm Register (BCNT3AR)

(1) In calendar count mode:

Address(es): RTC.RWKAR 0008 C416h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	DAYW[2:0]	Day-of-Week Setting	b2 b0 0 0 0: Sunday 0 0 1: Monday 0 1 0: Tuesday 0 1 1: Wednesday 1 0 0: Thursday 1 0 1: Friday 1 1 0: Saturday 1 1 1: Setting Prohibited	R/W
b6 to b3	—	Reserved	Set these bits to 0. They are read as the set value.	R/W
b7	ENB	ENB	0: The register value is not compared with the RWKCNT counter value. 1: The register value is compared with the RWKCNT counter value.	R/W

The RWKAR register is an alarm register corresponding to the coded day-of-week counter RWKCNT. When the ENB bit is set to 1, the RWKAR value is compared with the RWKCNT value. From among the alarm registers (RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, and RYRAREN), only those selected with the ENB bits set to 1 are compared with the corresponding counters. When the respective values all match, the IR flag corresponding to the ALM interrupt is set to 1.

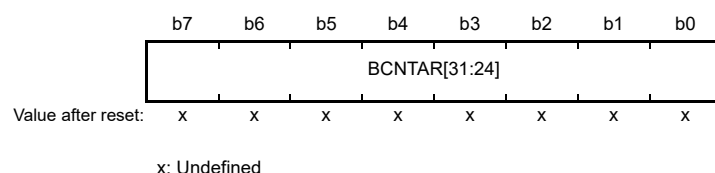
RWKAR values from 0 through 6 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly.

After writing to the RWKAR register, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

This register is set to 00h by an RTC software reset.

(2) In binary count mode:

Address(es): RTC.BCNT3AR 0008 C416h



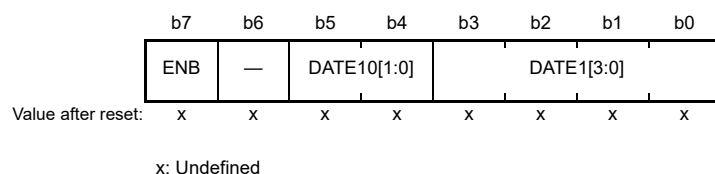
The BCNT3AR counter is a readable/writable alarm register corresponding to 32-bit binary counter b31 to b24.

This register is set to 00h by an RTC software reset.

33.2.13 Date Alarm Register (RDAYAR)/ Binary Counter 0 Alarm Enable Register (BCNT0AER)

(1) In calendar count mode:

Address(es): RTC.RDAYAR 0008 C418h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	DATE1[3:0]	1 Day	Value for the ones place of days	R/W
b5, b4	DATE10[1:0]	10 Days	Value for the tens place of days	R/W
b6	—	Reserved	Set this bit to 0. It is read as the set value.	R/W
b7	ENB	ENB	0: The register value is not compared with the RDAYCNT counter value. 1: The register value is compared with the RDAYCNT counter value.	R/W

The RDAYAR register is an alarm register corresponding to the BCD-coded date counter RDAYCNT. When the ENB bit is set to 1, the RDAYAR value is compared with the RDAYCNT value. From among the alarm registers (RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, and RYRAREN), only those selected with the ENB bits set to 1 are compared with the corresponding counters. When the respective values all match, the IR flag corresponding to the ALM interrupt is set to 1.

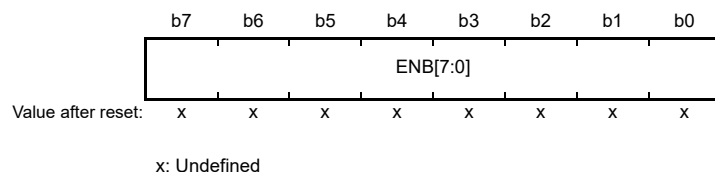
RDAYAR values from 01 through 31 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly.

After writing to the RDAYAR register, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

This register is set to 00h by an RTC software reset.

(2) In binary count mode:

Address(es): RTC.BCNT0AER 0008 C418h



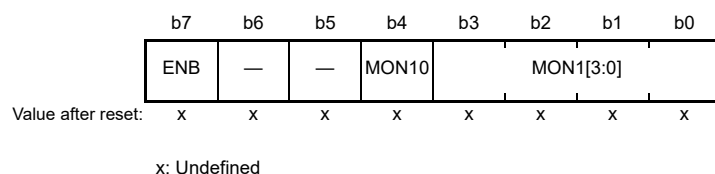
The BCNT0AER register is a readable/writable register for setting the alarm enable corresponding to 32-bit binary counter b7 to b0. Among the ENB[31:0] bits, the binary counter (BCNT[31:0]) corresponding to the bits which are set to 1 and the binary alarm register (BCNTAR[31:0]) are compared, and when all match, the IR flag corresponding to the ALM interrupt becomes 1.

This register is set to 00h by an RTC software reset.

33.2.14 Month Alarm Register (RMONAR)/ Binary Counter 1 Alarm Enable Register (BCNT1AER)

(1) In calendar count mode:

Address(es): RTC.RMONAR 0008 C41Ah



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	MON1[3:0]	1 Month	Value for the ones place of months	R/W
b4	MON10	10 Months	Value for the tens place of months	R/W
b6, b5	—	Reserved	Set these bits to 0. They are read as the set value.	R/W
b7	ENB	ENB	0: The register value is not compared with the RMONCNT counter value. 1: The register value is compared with the RMONCNT counter value.	R/W

The RMONAR register is an alarm register corresponding to the BCD-coded month counter RMONCNT. When the ENB bit is set to 1, the RMONAR value is compared with the RMONCNT value. From among the alarm registers (RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, and RYRAREN), only those selected with the ENB bits set to 1 are compared with the corresponding counters. When the respective values all match, the IR flag corresponding to the ALM interrupt is set to 1.

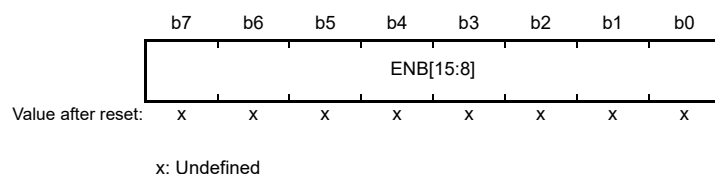
RMONAR values from 01 through 12 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly.

After writing to the RMONAR register, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

This register is set to 00h by an RTC software reset.

(2) In binary count mode:

Address(es): RTC.BCNT1AER 0008 C41Ah



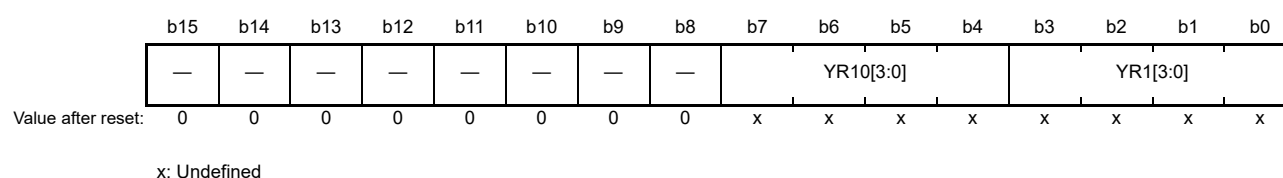
The BCNT1AER register is a readable/writable register for setting the alarm enable corresponding to 32-bit binary counter b15 to b8. Among the ENB[31:0] bits, the binary counter (BCNT[31:0]) corresponding to the bits which are set to 1 and the binary alarm register (BCNTAR[31:0]) are compared, and when all match, the IR flag corresponding to the ALM interrupt becomes 1.

This register is set to 00h by an RTC software reset.

33.2.15 Year Alarm Register (RYRAR)/ Binary Counter 2 Alarm Enable Register (BCNT2AER)

(1) In calendar count mode:

Address(es): RTC.RYRAR 0008 C41Ch



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	YR1[3:0]	1 Year	Value for the ones place of years	R/W
b7 to b4	YR10[3:0]	10 Years	Value for the tens place of years	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The RYRAR register is an alarm register corresponding to the BCD-coded year counter RYRCNT.

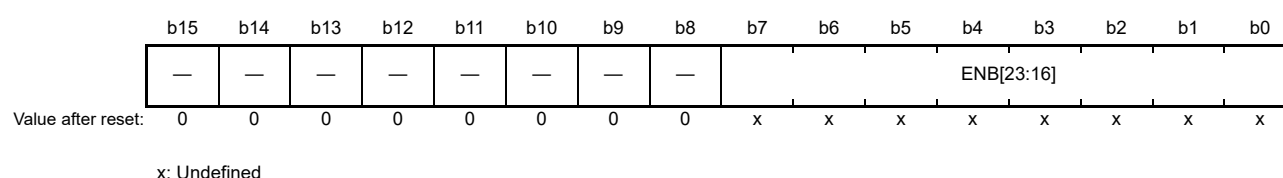
RYRAR values from 00 through 99 (in BCD) can be specified; if a value outside of this range is specified, the RTC does not operate correctly.

After writing to the RYRAR register, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers for notes on accessing registers.

This register is set to 0000h by an RTC software reset.

(2) In binary count mode:

Address(es): RTC.BCNT2AER 0008 C41Ch



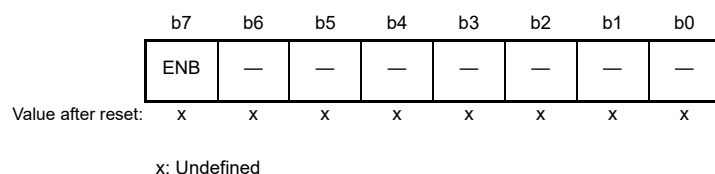
The BCNT2AER register is a readable/writable register for setting the alarm enable corresponding to 32-bit binary counter b23 to b16. Among the ENB[31:0] bits, the binary counter (BCNT[31:0]) corresponding to the bits which are set to 1 and the binary alarm register (BCNTAR[31:0]) are compared, and when all match, the IR flag corresponding to the ALM interrupt becomes 1.

This register is set to 0000h by an RTC software reset.

33.2.16 Year Alarm Enable Register (RYRAREN)/ Binary Counter 3 Alarm Enable Register (BCNT3AER)

(1) In calendar count mode:

Address(es): RTC.RYRAREN 0008 C41Eh



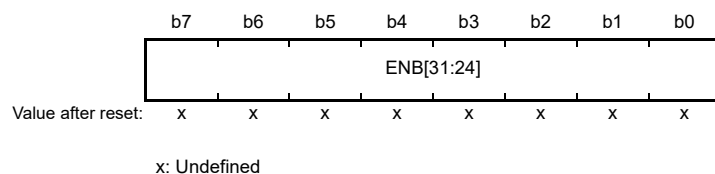
Bit	Symbol	Bit Name	Description	R/W
b6 to b0	—	Reserved	Set these bits to 0. They are read as the set value.	R/W
b7	ENB	ENB	0: The register value is not compared with the RYRCNT counter value. 1: The register value is compared with the RYRCNT counter value.	R/W

When the ENB bit in the RYRAREN register is set to 1, the RYRAR value is compared with the RYRCNT value. From among the alarm registers (RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, and RYRAREN), only those selected with the ENB bits set to 1 are compared with the corresponding counters. When the respective values all match, the IR flag corresponding to the ALM interrupt is set to 1.

This register is set to 00h by an RTC software reset.

(2) In binary count mode:

Address(es): RTC.BCNT3AER 0008 C41Eh

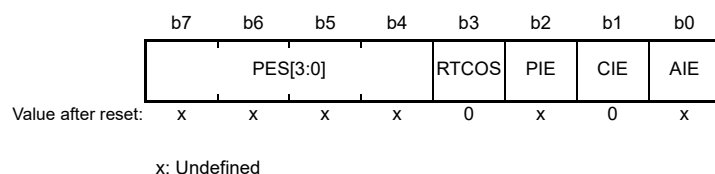


The BCNT3AER register is a readable/writable register for setting the alarm enable corresponding to 32-bit binary counter b31 to b24. Among the ENB[31:0] bits, the binary counter (BCNT[31:0]) corresponding to the bits which are set to 1 and the binary alarm register (BCNTAR[31:0]) are compared, and when all match, the IR flag corresponding to the ALM interrupt becomes 1.

This register is set to 00h by an RTC software reset.

33.2.17 RTC Control Register 1 (RCR1)

Address(es): RTC.RCR1 0008 C422h



Bit	Symbol	Bit Name	Description	R/W
b0	AIE	Alarm Interrupt Enable	0: An alarm interrupt request is disabled. 1: An alarm interrupt request is enabled.	R/W
b1	CIE	Carry Interrupt Enable	0: A carry interrupt request is disabled. 1: A carry interrupt request is enabled.	R/W
b2	PIE	Periodic Interrupt Enable	0: A periodic interrupt request is disabled. 1: A periodic interrupt request is enabled.	R/W
b3	RTCOS	RTCOUT Output Select	0: RTCOUT outputs 1 Hz. 1: RTCOUT outputs 64 Hz.	R/W
b7 to b4	PES[3:0]	Periodic Interrupt Select	b7 b4 0 1 1 0: A periodic interrupt is generated every 1/256 second.*1 0 1 1 1: A periodic interrupt is generated every 1/128 second. 1 0 0 0: A periodic interrupt is generated every 1/64 second. 1 0 0 1: A periodic interrupt is generated every 1/32 second. 1 0 1 0: A periodic interrupt is generated every 1/16 second. 1 0 1 1: A periodic interrupt is generated every 1/8 second. 1 1 0 0: A periodic interrupt is generated every 1/4 second. 1 1 0 1: A periodic interrupt is generated every 1/2 second. 1 1 1 0: A periodic interrupt is generated every 1 second. 1 1 1 1: A periodic interrupt is generated every 2 seconds. Other than above: No periodic interrupts are generated.	R/W

Note 1. When the main clock is selected (RCR4.RCKSEL bit = 1) while the PES[3:0] bits are 0110b, a periodic interrupt is generated every 1/128 second.

The RCR1 register is used in both calendar count mode and in binary count mode.

Bits AIE, PIE, and PES[3:0] are updated synchronously with the count source. When the RCR1 register is modified, check that all the bits have been updated before proceeding to the next processing.

AIE Bit (Alarm Interrupt Enable)

This bit enables or disables alarm interrupt requests.

If the times indicated by the counters and alarm settings match in deep software standby mode, the MCU returns from the mode regardless of the value of the AIE bit.

CIE Bit (Carry Interrupt Enable)

This bit enables and disables interrupt requests when a carry to the RSECCNT/BCNT0 register occurs, or when a carry to the 64-Hz counter (R64CNT) occurs while reading the 64-Hz counter.

PIE Bit (Periodic Interrupt Enable)

This bit enables or disabled a periodic interrupt.

If the periods indicated by the counters and PES[3:0] settings match in deep software standby mode, the MCU returns from the mode regardless of the value of the PIE bit.

RTCOS Bit (RTCOUT Output Select)

This bit selects the RTCOUT output period. The RTCOS bit must be rewritten while count operation is stopped (the RCR2.START bit is 0) and RTCOUT output is disabled (the RCR2.RTCOE bit is 0). When the RTCOUT is output to an external pin, the RCR2.RTCOE bit must be enabled. For details on controlling I/O ports, refer to **section 23.4.1, Procedure for Specifying Input/Output Pin Function**.

PES[3:0] Bits (Periodic Interrupt Select)

These bits specify the period for the periodic interrupt. A periodic interrupt is generated with the period specified by these bits.

33.2.18 RTC Control Register 2 (RCR2)

Address(es): RTC.RCR2 0008 C424h

	b7	b6	b5	b4	b3	b2	b1	b0
	CNTMD	HR24	AADJP	AADJE	RTCOE	ADJ30	RESET	START
Value after reset:	x	x	x	x	0	0	0	x

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	START	Start*4	0: Prescaler and counter are stopped. 1: Prescaler and counter operate normally.	R/W
b1	RESET	RTC Software Reset	- In writing 0: Writing is invalid. 1: The prescaler and the target registers for RTC software reset*1 are initialized. - In reading 0: In normal time operation, or an RTC software reset has completed. 1: During an RTC software reset	R/W
b2	ADJ30	30-Second Adjustment*2	- In writing 0: Writing is invalid. 1: 30-second adjustment is executed. - In reading 0: In normal time operation, or 30-second adjustment has completed. 1: During 30-second adjustment	R/W
b3	RTCOE	RTCOUT Output Enable	0: RTCOUT output disabled. 1: RTCOUT output enabled.	R/W
b4	AADJE	Automatic Adjustment Enable*3, *4	0: Automatic adjustment is disabled. 1: Automatic adjustment is enabled.	R/W
b5	AADJP	Automatic Adjustment Period Select*3, *4	0: The RADJ.ADJ[5:0] setting value is adjusted from the count value of the prescaler every minute (every 32 seconds in binary counter mode). 1: The RADJ.ADJ[5:0] setting value is adjusted from the count value of the prescaler every 10 seconds (every 8 seconds in binary counter mode).	R/W
b6	HR24	Hours Mode*2, *4	0: The RTC operates in 12-hour mode. 1: The RTC operates in 24-hour mode.	R/W
b7	CNTMD	Count Mode Select*4	0: The calendar count mode. 1: The binary count mode.	R/W

Note 1. R64CNT, RSECAR/BCNT0AR, RMINAR/BCNT1AR, RHRAR/BCNT2AR, RWKAR/BCNT3AR, RDAYAR/BCNT0AER, RMONAR/BCNT1AER, RYRAR/BCNT2AER, RYRAREN/BCNT3AER, RADJ, RTCCRn, RSECCPn/BCNT0CPn, RMINCPn/BCNT1CPn, RHRCpPn/BCNT2CPn, RDAYCPn/BCNT3CPn, RMONCPn, RCR2.ADJ30, RCR2.AADJE, RCR2.AADJP

Note 2. This bit is reserved in binary counter mode. The write value should be 0.

Note 3. When the main clock is selected, the setting of this bit is disabled.

Note 4. After writing to this bit, confirm that its value has actually changed before proceeding with further processing. Refer to section 33.6.5, Notes on Writing to and Reading from Registers regarding changes to the values of the AADJE, AADJP, and HR24 bits.

The RCR2 register is related to hours mode, automatic adjustment function, enabling the RTCOUT output, 30-second adjustment, RTC software reset, and controlling count operation.

START Bit (Start)

This bit stops or restarts the prescaler or counter (clock) operation.

The START bit is updated in synchronization with the count source. When the START bit is modified, check that the bit is updated before proceeding to the next processing.

RESET Bit (RTC Software Reset)

This bit initializes the prescaler and registers to be reset by RTC software reset.

When 1 is written to the RESET bit, the initialization starts in synchronization with the count source. When the initialization is completed, the RESET bit is automatically set to 0.

When 1 is written to the RESET bit, check that the bit is set to 0, and then make next settings.

ADJ30 Bit (30-Second Adjustment)

This bit is for 30-second adjustment.

When 1 is written to the ADJ30 bit, the RSECCNT value of 30 seconds or less is rounded down to 00 second and the value of 30 seconds or more is rounded up to 1 minute.

The 30-second adjustment is performed in synchronization with the count source. When 1 is written to this bit, the ADJ30 bit is automatically set to 0 after the 30-second adjustment is completed. In case when 1 is written to the ADJ30 bit, check that the bit is set to 0, and then make next settings.

When the 30-second adjustment is performed, the prescaler and R64CNT are also reset.

The ADJ30 bit is set to 0 by an RTC software reset.

This bit is reserved in binary counter mode. The write value should be 0.

RTCOE Bit (RTCOUT Output Enable)

This bit enables output of a 1-Hz/64-Hz clock signal from the RTCOUT pin.

Use the START bit to stop counting by the counters before changing the value of the RTCOE bit. Do not stop counting (write 0 to the START bit) and change the value of the RTCOE bit at the same time.

When RTCOUT signal is to be output from an external pin, set the RTCOE bit to 1 and set up the port control for the pin.

AADJE Bit (Automatic Adjustment Enable)

This bit controls (enables or disables) automatic adjustment.

Set the plus-minus bits (RADJ.PMADJ[1:0]) to 00b (adjustment is not performed) before changing the value of the AADJE bit.

The AADJE bit is set to 0 by an RTC software reset.

AADJP Bit (Automatic Adjustment Period Select)

This bit selects the automatic-adjustment period.

Set the plus-minus bits (RADJ.PMADJ[1:0]) to 00b (adjustment is not performed) before changing the value of the AADJP bit.

The AADJP bit is set to 0 by an RTC software reset.

HR24 Bit (Hours Mode)

This bit specifies whether the RTC will operate in 12- or 24-hour mode.

Use the START bit to stop counting by the counters before changing the value of the HR24 bit. Do not stop counting (write 0 to the START bit) and change the value of the HR24 bit at the same time.

This bit is reserved in binary counter mode. The write value should be 0.

CNTMD Bit (Count Mode Select)

This bit specifies whether the RTC count mode is operated in calendar count mode or in binary count mode.

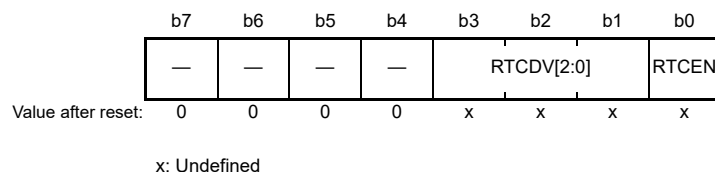
After setting the count mode, execute an RTC software reset and start again from the initial settings.

The CNTMD bit is updated in synchronization with the count source, so when the value of the CNTMD bit has been changed, check that the value of the bit has actually been updated before applying the RTC software reset. The count mode changes to that which was specified beforehand in the CNTMD bit after the RTC software reset is applied.

For details on initial settings, refer to section 33.3.1, Outline of Initial Settings of Registers after Power On.

33.2.19 RTC Control Register 3 (RCR3)

Address(es): RTC.RCR3 0008 C426h



Bit	Symbol	Bit Name	Description	R/W
b0	RTCEN	Sub-Clock Oscillator Control	0: Sub-clock oscillator is stopped. 1: Sub-clock oscillator is operating.	R/W
b3 to b1	RTCDV[2:0]	Sub-Clock Oscillator Drive Capacity Control	b3 b1 0 0 1: Drive capacity for low CL 1 1 0: Drive capacity for standard CL Settings other than above are prohibited.	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The RCR3 register is used for controlling the sub-clock oscillator in the clock generation circuit. For details on controlling the sub-clock oscillator, refer to [section 9, Clock Generation Circuit](#).

This register is a function common to calendar count mode and binary count mode.

When this register is modified, check that all the bits have been updated before proceeding to the next processing.

RTCEN Bit (Sub-Clock Oscillator Control)

The RTCEN bit and a clock generation circuit register (the SOSCCR.SOSTP bit) control whether to operate or stop the sub-clock oscillator. If one of the bits is set so as to enable the operation, the sub-clock oscillator runs.

When using the sub-clock as the count source to the RTC, set the sub-clock oscillator using the RTCEN bit.

When the main clock is selected (RCR4.RCKSEL bit = 1), making the sub-clock oscillator run or stop is controlled only by the SOSCCR.SOSTP bit regardless of the value set in the RTCEN bit.

RTCDV[2:0] Bits (Sub-Clock Oscillator Drive Capacity Control)

These bits control the drive capacity of the sub-clock oscillator.

Before modifying the RTCDV[2:0] bits, be sure to stop the sub-clock oscillator. When the main clock is selected (RCR4.RCKSEL bit = 1), making the sub-clock oscillator run or stop is controlled only by the SOSCCR.SOSTP bit regardless of the value set in the RTCEN bit.

33.2.19.1 Notes on using a low CL crystal unit

When the RCR3.RTCDV[2:0] bits are 001b (drive capacity for low CL), the oscillator is susceptible to noise. Especially when the signal level of any pin near the XCIN or XCOUT pin is changed, the oscillation accuracy of the sub-clock oscillator may be affected. The accuracy is affected differently depending on the board traces and how the signal level of any pin near the XCIN or XCOUT pin is changed. When designing a board using a low CL crystal unit, refer to the application note “Design Guide for Low CL Sub-clock Circuits” (R01AN1187EJ) to reduce the influence from noise. The following are examples that may significantly affect oscillation accuracy:

(1) When connecting an on-chip debugging emulator to the FINED pin

Since the FINED pin (FINE interface pin) is near the XCIN and XCOUT pins, the oscillation accuracy of the sub-clock oscillator is affected when using the FINED pin in debugging. When using the FINED pin in debugging, set the RCR3.RTCDV[2:0] bits to 001b (drive capacity for low CL) and debug at the room temperature.

(2) When supplying an external clock to the main clock oscillator

When inputting an external clock to the EXTAL pin, the oscillation accuracy of the sub-clock oscillator may be affected. When inputting an inverted external clock to the XTAL pin, the oscillation accuracy will be affected more significantly.

33.2.20 RTC Control Register 4 (RCR4)

Address(es): RTC.RCR4 0008 C428h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	RCKSEL
0	0	0	0	0	0	0	x

Value after reset:

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	RCKSEL	Count Source Select	0: Sub-clock oscillator is selected. 1: Main clock oscillator is selected.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The RCR4 register is used for selecting the count source. This function is used in both calendar count mode and binary count mode.

When the RCKSEL bit is set to 0, the time is counted with the sub-clock. When the bit is set to 1, the time is counted with the main clock.

RCKSEL Bit (Count Source Select)

This bit selects the count source from the sub-clock and the main clock.

The count source should be selected only once before the initial settings of the RTC registers at power on.

33.2.21 Frequency Register H/L (RFRH/RFRL)

Address(es): RTC.RFRH 0008 C42Ah

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RFC[16]
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x

Value after reset:

x: Undefined

Address(es): RTC.RFRL 0008 C42Ch

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
RFC[15:0]															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Value after reset:

x: Undefined

- RFRH register

Bit	Symbol	Bit Name	Description	R/W
b0	RFC[16]	Frequency Divide Ratio Setting	Set the divide ratio of the main clock to generate the reference clock of 128 Hz. If the setting value in the RFC[16:0] bits is n, the prescaler divides the main clock by n + 1.	R/W
b15 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

- RFRL register

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	RFC[15:0]	Frequency Divide Ratio Setting	Set the divide ratio of the main clock to generate the reference clock of 128 Hz. If the setting value in the RFC[16:0] bits is n, the prescaler divides the main clock by n + 1.	R/W

RFRH/RFRL is a register for controlling the prescaler when the main clock is selected.

The RTC time counter operates on a 128-Hz clock signal as the reference clock. Therefore, when the main clock is selected, the main clock is divided by the prescaler to generate a 128-Hz clock signal. Set the frequency divide ratio in the RFC[16:0] bits to generate a 128-Hz clock from the main clock frequency. As for the calculation method, refer to the formula below.

A value from 0 0007h through 1 FFFFh can be specified in the RFC[16:0] bits; if a value outside of this range is specified, the RTC does not operate correctly. Rewrite this register while the RCR2.START bit is 0 (counter is stopped). In addition, the range of the main clock frequency that can be used to generate a 128-Hz clock is 1.024 kHz to 16.778 MHz.

The frequency of the peripheral module clock and the main clock should be in the relationship that the peripheral module clock $\geq$ the main clock.

- Calculation method of frequency comparison value

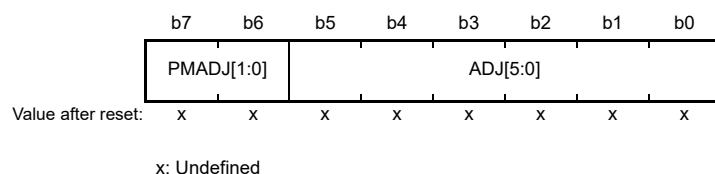
$$\text{RFC}[16:0] = (\text{Main clock frequency}) \div 128 - 1$$

Table 33.3 RFRH/RFRL Register Settings by the Main Clock Frequency

Main Clock Frequency	RFRH/RFRL Register Setting Value
4 MHz	0000 7A11h
8 MHz	0000 F423h
10 MHz	0001 312Ch
12 MHz	0001 6E35h
16 MHz	0001 E847h

33.2.22 Time Error Adjustment Register (RADJ)

Address(es): RTC.RADJ 0008 C42Eh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	ADJ[5:0]	Adjustment Value	These bits specify the adjustment value from the prescaler.	R/W
b7, b6	PMADJ[1:0]	Plus-Minus	b7 b6 0 0: Adjustment is not performed. 0 1: Adjustment is performed by the addition to the prescaler. 1 0: Adjustment is performed by the subtraction from the prescaler. 1 1: Setting prohibited	R/W

Adjustment is performed by the addition to or subtraction from the prescaler.

In case when the automatic adjustment enable (RCR2.AADJE) bit is 0, adjustment is performed when writing to the RADJ.

In case when the RCR2.AADJE bit is 1, adjustment is performed in the interval specified by the automatic adjustment period select (RCR2.AADJP) bit.

The current adjustment by software (disabling automatic adjustment) may be invalid if the following adjustment value is specified within 320 cycles of the count source after the register setting. To perform adjustment consecutively, wait for 320 cycles or more of the count source after the register setting and then specify the next adjustment value.

RADJ is updated in synchronization with the count source. When RADJ is modified, check that all the bits have been updated before continuing with further processing.

This register is set to 00h by an RTC software reset.

The setting of this register is enabled only when the sub-clock is selected.

When the main clock is selected, adjustment is not performed.

ADJ[5:0] Bits (Adjustment Value)

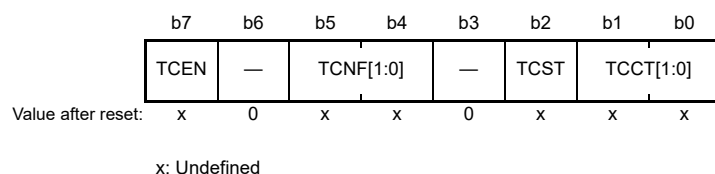
These bits specify the adjustment value (the number of sub-clock cycles) from the prescaler.

PMADJ[1:0] Bits (Plus-Minus)

These bits select whether the clock is set ahead or back depending on the error-adjustment value set in the ADJ[5:0] bits.

33.2.23 Time Capture Control Register n (RTCCRN) (n = 0 to 2)

Address(es): RTC.RTCCR0 0008 C440h, RTC.RTCCR1 0008 C442h, RTC.RTCCR2 0008 C444h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	TCCT[1:0]	Time Capture Control	b1 b0 0 0: No event is detected. 0 1: Rising edge is detected. 1 0: Falling edge is detected. 1 1: Both edges are detected.	R/W
b2	TCST	Time Capture Status Flag	0: No event is detected. 1: An event is detected.	R/(W) *1
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5, b4	TCNF[1:0]	Time Capture Noise Filter Control	b5 b4 0 0: The noise filter is off. 0 1: Setting prohibited 1 0: The noise filter is on (count source). 1 1: The noise filter is on (count source by divided by 32).	R/W
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	TCEN	Time Capture Event Input Pin Enable	0: The RTCICn pin is disabled as the time capture event input. 1: The RTCICn pin is enabled as the time capture event input.	R/W

Note 1. Writing 1 has no effect. This flag is cleared by writing 0.

The RTCCRN register is used both in calendar count mode and in binary count mode.

RTCCR0, RTCCR1, and RTCCR2 control the RTCIC0, RTCIC1, and RTCIC2 pins, respectively.

RTCCRN is updated in synchronization with the count source. When RTCCRN is modified, check that all the bits except for the TCST flag have been updated before continuing with further processing.

This register is set to 00h by an RTC software reset.

TCCT[1:0] Bits (Time Capture Control)

These bits control the edge detection of the time capture event input pins (RTCIC0, RTCIC1, and RTCIC2). The detection edge is selectable. The TCCT[1:0] bits should be set while the TCEN bit is 1.

TCST Flag (Time Capture Status Flag)

This flag indicates that an event of the time capture event input pins (RTCIC0, RTCIC1, and RTCIC2) has been detected. When the TCST flag is 0, no event is detected.

When the TCST flag is 1, this flag indicates that an event of the corresponding pin has been detected and the capture register is valid. When multiple events have been detected, the capture time for the first event is retained. If an event is detected while the count operation is stopped (the RCR2.START bit is 0), the captured value is not guaranteed. In this case, set the TCST flag to 0 for deleting the captured value.

Writing 0 sets the TCST flag to 0. In addition, writing any other value except 0 has no effect.

Set the TCST flag while the TCCT[1:0] bits are 00b (no event is detected).

The TCST flag is set to 0 in synchronization with the count source. When the TCST flag is set to 0, check that the flag has been updated before continuing with further processing.

TCNF[1:0] Bits (Time Capture Noise Filter Control)

These bits control the noise filter of the time capture event input pins (RTCIC0, RTCIC1, and RTCIC2).

When the noise filter is on, the count source divided by 1 or divided by 32 is selectable. In this case, when the input level on the time capture event input pin matches three consecutive times at the set sampling period, the input level is determined.

Set the TCNF[1:0] bits while the TCCT[1:0] bits are 00b (no event is detected). When the noise filter is used, set the TCEN bit to 1, set the TCNF[1:0] bits, wait for three cycles of the specified sampling period, and then set the TCCT[1:0] bits.

TCEN Bit (Time Capture Event Input Pin Enable)

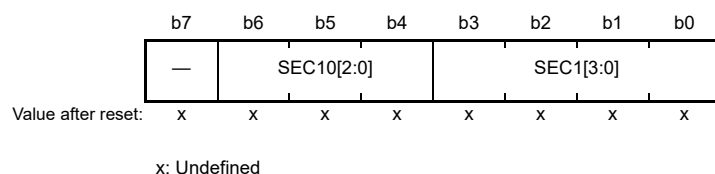
This bit enables or disables the time capture event input pins (RTCIC0, RTCIC1, and RTCIC2).

When the sub-clock is selected (RCR4.RCKSEL bit = 0) and the sub-clock oscillator is stopped (RCR3.RTCEN bit = 0), the time capture event input pins (RTCIC0, RTCIC1, and RTCIC2) are disabled regardless of the value of the TCEN bit. When the time capture event input pin functions (RTCIC0, RTCIC1, and RTCIC2) are allocated to the same pins as other multiplexed functions, make the general input port pin settings for the corresponding pins and then set this bit to 1. If the TCEN bit is set to 0, set also the TCCT[1:0] bits to 00b.

33.2.24 Second Capture Register n (RSECCPn) (n = 0 to 2)/ BCNT0 Capture Register n (BCNT0CPn) (n = 0 to 2)

(1) In calendar count mode:

Address(es): RTC.RSECCP0 0008 C452h, RTC.RSECCP1 0008 C462h, RTC.RSECCP2 0008 C472h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	SEC1[3:0]	1-Second Capture	Capture value for the ones place of seconds	R
b6 to b4	SEC10[2:0]	10-Second Capture	Capture value for the tens place of seconds	R
b7	—	Reserved	This bit is read as 0 after an RTC software reset.	R

RSECCPn is a read-only register that captures the RSECCNT value when a time capture event is detected.

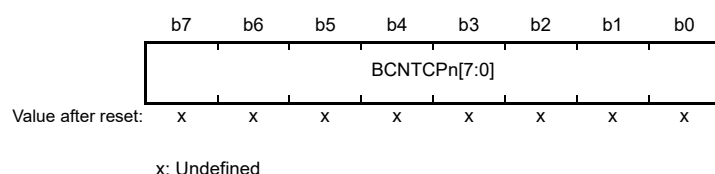
The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RSECCP0, RSECCP1, and RSECCP2 registers, respectively.

This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

(2) In binary count mode:

Address(es): RTC.BCNT0CP0 0008 C452h, RTC.BCNT0CP1 0008 C462h, RTC.BCNT0CP2 0008 C472h



BCNT0CPn is a read-only register that captures the BCNT0 value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the BCNT0CP0, BCNT0CP1, and BCNT0CP2 registers, respectively.

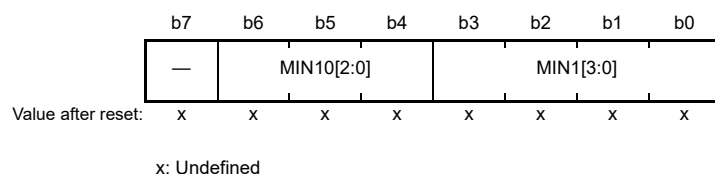
This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

33.2.25 Minute Capture Register n (RMINCPn) (n = 0 to 2)/ BCNT1 Capture Register n (BCNT1CPn) (n = 0 to 2)

(1) In calendar count mode:

Address(es): RTC.RMINCP0 0008 C454h, RTC.RMINCP1 0008 C464h, RTC.RMINCP2 0008 C474h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	MIN1[3:0]	1-Minute Capture	Capture value for the ones place of minutes	R
b6 to b4	MIN10[2:0]	10-Minute Capture	Capture value for the tens place of minutes	R
b7	—	Reserved	This bit is read as 0 after an RTC software reset.	R

RMINCPn is a read-only register that captures the RMINCNT value when a time capture event is detected.

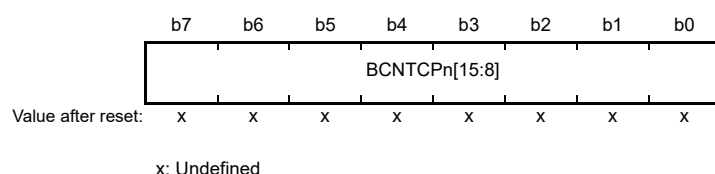
The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RMINCP0, RMINCP1, and RMINCP2 registers, respectively.

This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

(2) In binary count mode:

Address(es): RTC.BCNT1CP0 0008 C454h, RTC.BCNT1CP1 0008 C464h, RTC.BCNT1CP2 0008 C474h



BCNT1CPn is a read-only register that captures the BCNT1 value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the BCNT1CP0, BCNT1CP1, and BCNT1CP2 registers, respectively.

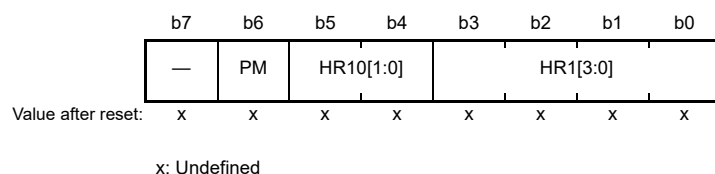
This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

33.2.26 Hour Capture Register n (RHRCp_n) (n = 0 to 2)/ BCNT2 Capture Register n (BCNT2CP_n) (n = 0 to 2)

(1) In calendar count mode:

Address(es): RTC.RHRCp0 0008 C456h, RTC.RHRCp1 0008 C466h, RTC.RHRCp2 0008 C476h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	HR1[3:0]	1-Hour Capture	Capture value for the ones place of hours	R
b5, b4	HR10[1:0]	10-Hour Capture	Capture value for the tens place of hours	R
b6	PM	PM	0: a.m. 1: p.m.	R
b7	—	Reserved	This bit is read as 0 after an RTC software reset.	R

RHRCp_n is a read-only register that captures the RHRCNT value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RHRCp0, RHRCp1, and RHRCp2 registers, respectively.

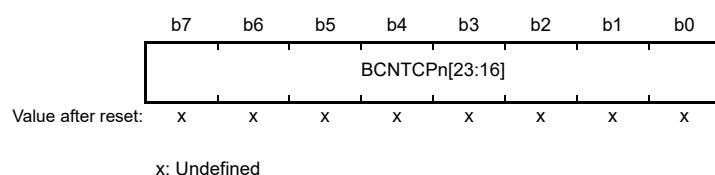
The PM bit is only enabled when the RCR2.HR24 bit is 0 (in 12-hour mode).

This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

(2) In binary count mode:

Address(es): RTC.BCNT2CP0 0008 C456h, RTC.BCNT2CP1 0008 C466h, RTC.BCNT2CP2 0008 C476h



BCNT2CP_n is a read-only register that captures the BCNT2 value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the BCNT2CP0, BCNT2CP1, and BCNT2CP2 registers, respectively.

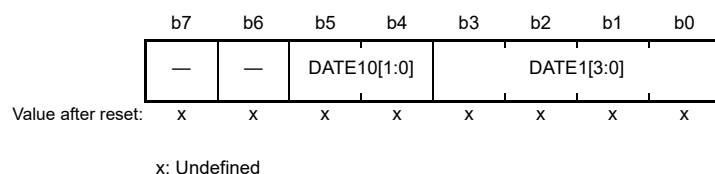
This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

33.2.27 Date Capture Register n (RDAYCPn) (n = 0 to 2)/ BCNT3 Capture Register n (BCNT3CPn) (n = 0 to 2)

(1) In calendar count mode:

Address(es): RTC.RDAYCP0 0008 C45Ah, RTC.RDAYCP1 0008 C46Ah, RTC.RDAYCP2 0008 C47Ah



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	DATE1[3:0]	1-Day Capture	Capture value for the ones place of days	R
b5, b4	DATE10[1:0]	10-Day Capture	Capture value for the tens place of days	R
b7, b6	—	Reserved	These bits are read as 0 after an RTC software reset.	R

RDAYCPn is a read-only register that captures the RDAYCNT value when a time capture event is detected.

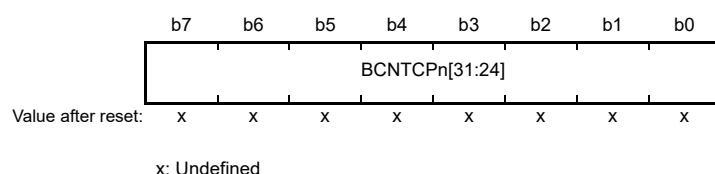
The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RDAYCP0, RDAYCP1, and RDAYCP2 registers, respectively.

This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

(2) In binary count mode:

Address(es): RTC.BCNT3CP0 0008 C45Ah, RTC.BCNT3CP1 0008 C46Ah, RTC.BCNT3CP2 0008 C47Ah



BCNT3CPn is a read-only register that captures the BCNT3 value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the BCNT3CP0, BCNT3CP1, and BCNT3CP2 registers, respectively.

This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

33.2.28 Month Capture Register n (RMONCPn) (n = 0 to 2)

Address(es): RTC.RMONCP0 0008 C45Ch, RTC.RMONCP1 0008 C46Ch, RTC.RMONCP2 0008 C47Ch

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	MON10	MON1[3:0]			
Value after reset:	0	0	0	x	x	x	x	x

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	MON1[3:0]	1-Month Capture	Capture value for the ones place of months	R
b4	MON10	10-Month Capture	Capture value for the tens place of months	R
b7 to b5	—	Reserved	These bits are read as 0	R

RMONCPn is a read-only register that captures the RMONCNT value when a time capture event is detected. The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RMONCP0, RMONCP1, and RMONCP2 registers, respectively.

This register is set to 00h by an RTC software reset.

Before reading from this register, be sure to stop the time capture event detection through the setting of the RTCCRn.TCCT[1:0] bits.

33.3 Operation

33.3.1 Outline of Initial Settings of Registers after Power On

After the power is turned on, the initial settings for the clock setting, count mode setting, time error adjustment, time setting, alarm, interrupt, and time capture control register should be performed.

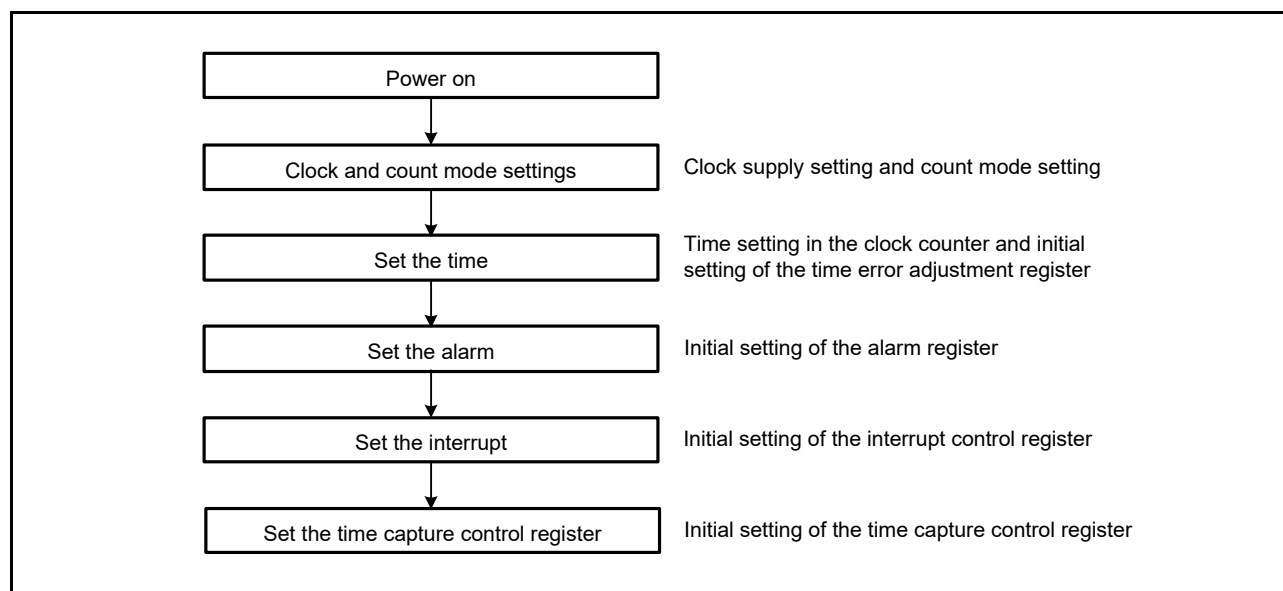


Figure 33.2 Outline of Initial Settings after Power On

33.3.2 Clock and Count Mode Setting Procedure

Figure 33.3 shows how to set the clock and the count mode.

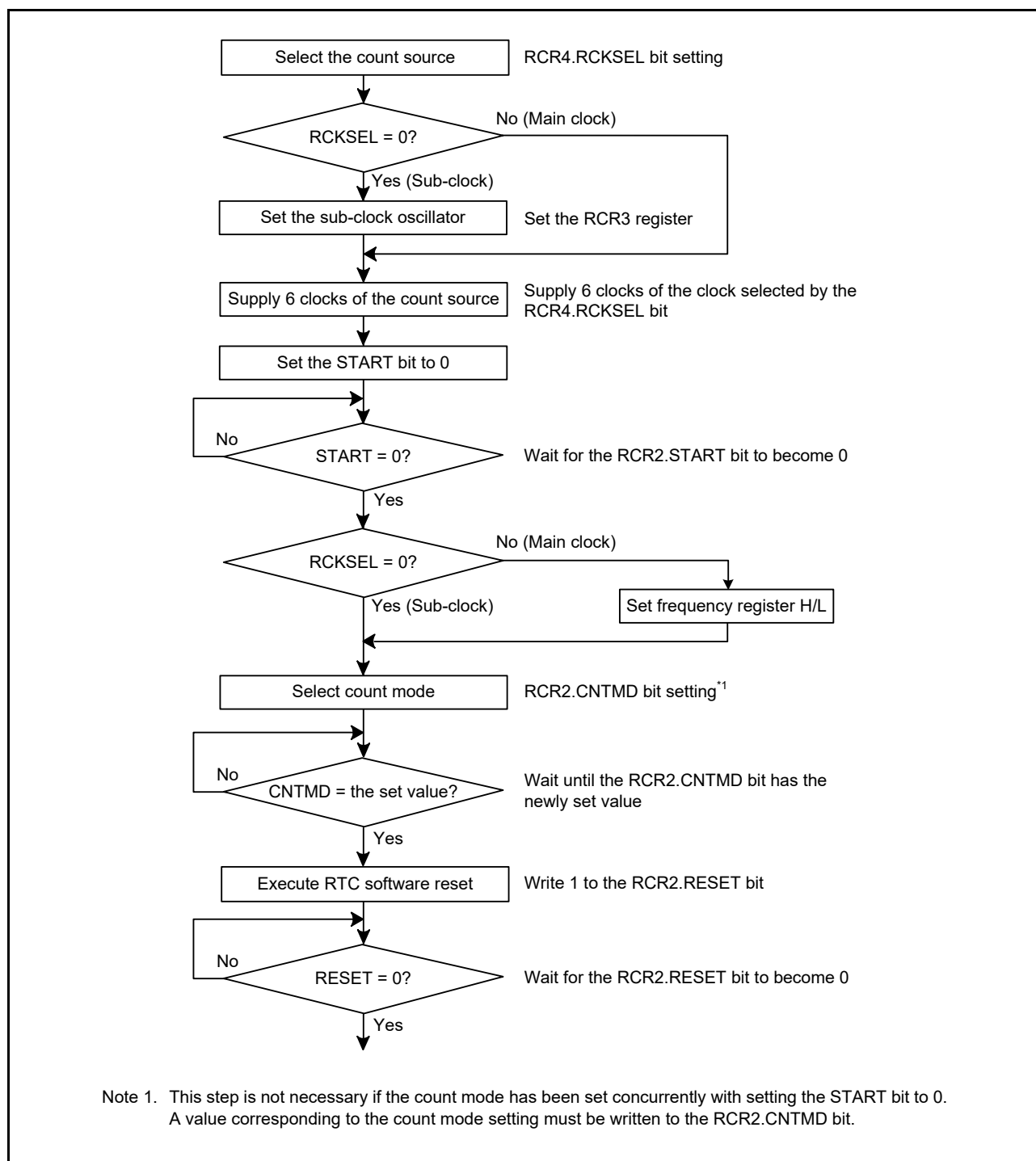


Figure 33.3 Clock and Count Mode Setting Procedure

33.3.3 Setting the Time

Figure 33.4 shows how to set the time.

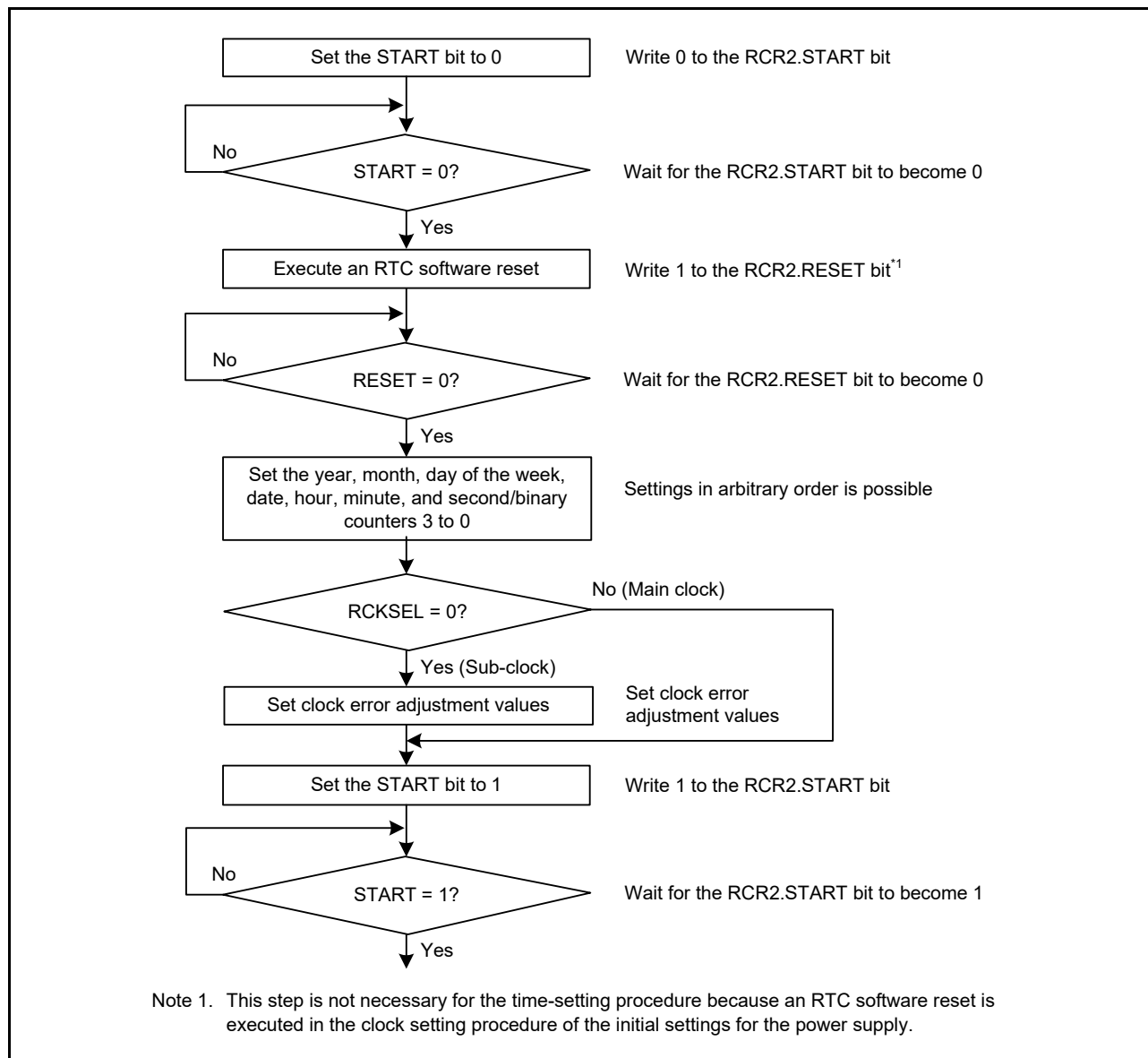


Figure 33.4 Setting the Time

33.3.4 30-Second Adjustment

Figure 33.5 shows how to execute 30-second adjustment.

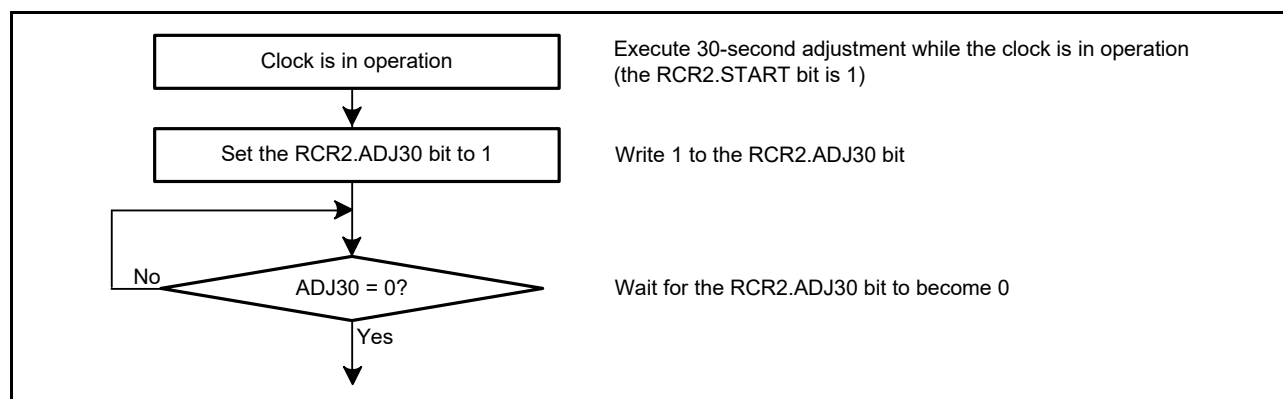


Figure 33.5 30-Second Adjustment

33.3.5 Reading 64-Hz Counter and Time

Figure 33.6 shows how to read the 64-Hz counter and time.

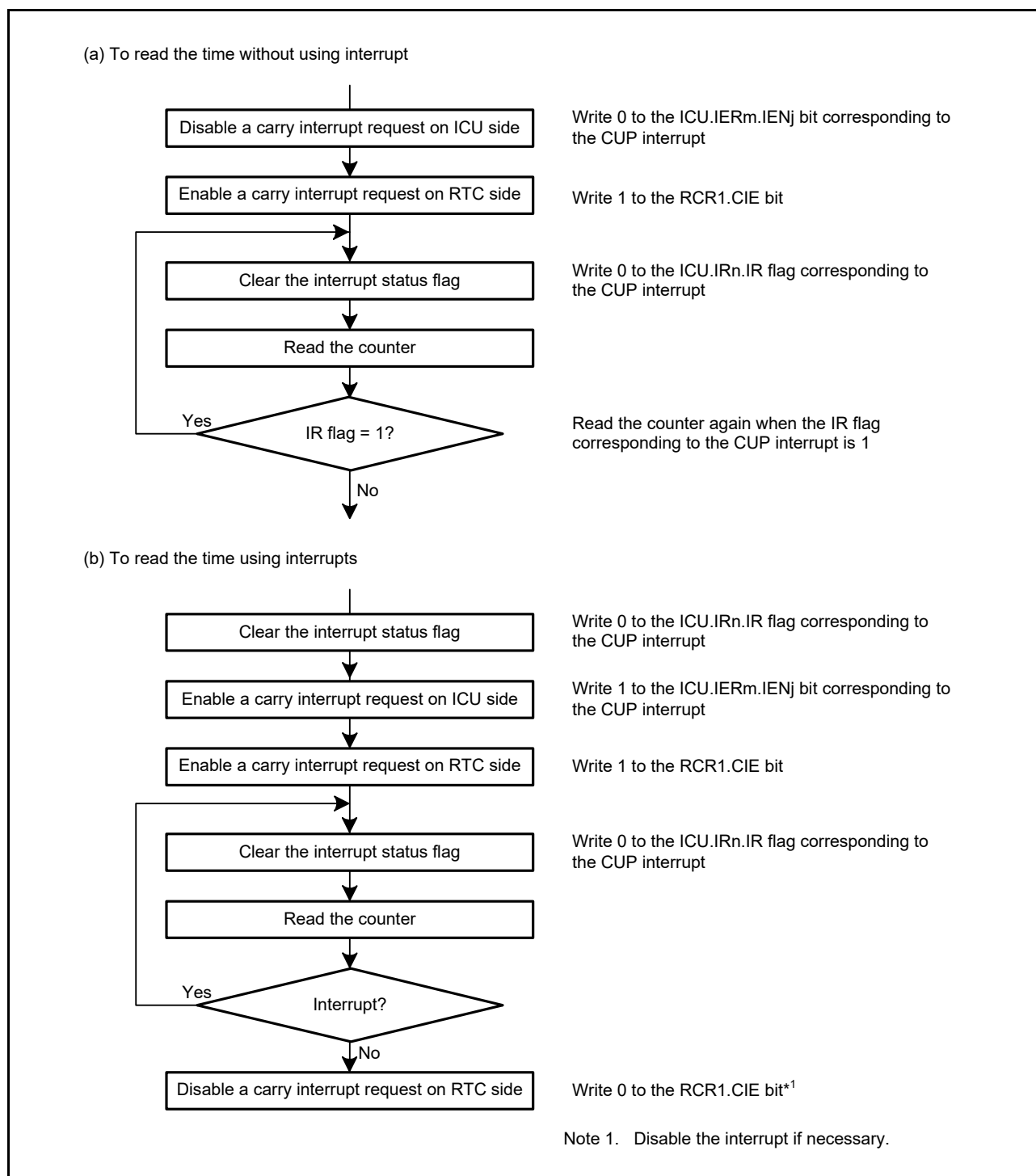


Figure 33.6 Reading Time

If a carry occurs while the 64-Hz counter and time are being read, the correct time will not be obtained, so they must be read again. The procedure for reading the time without using interrupts is shown in (a) in Figure 33.6, and the procedure using carry interrupts in (b). To keep the program simple, method (a) should be used in most cases.

33.3.6 Alarm Function

Figure 33.7 shows how to use the alarm function.

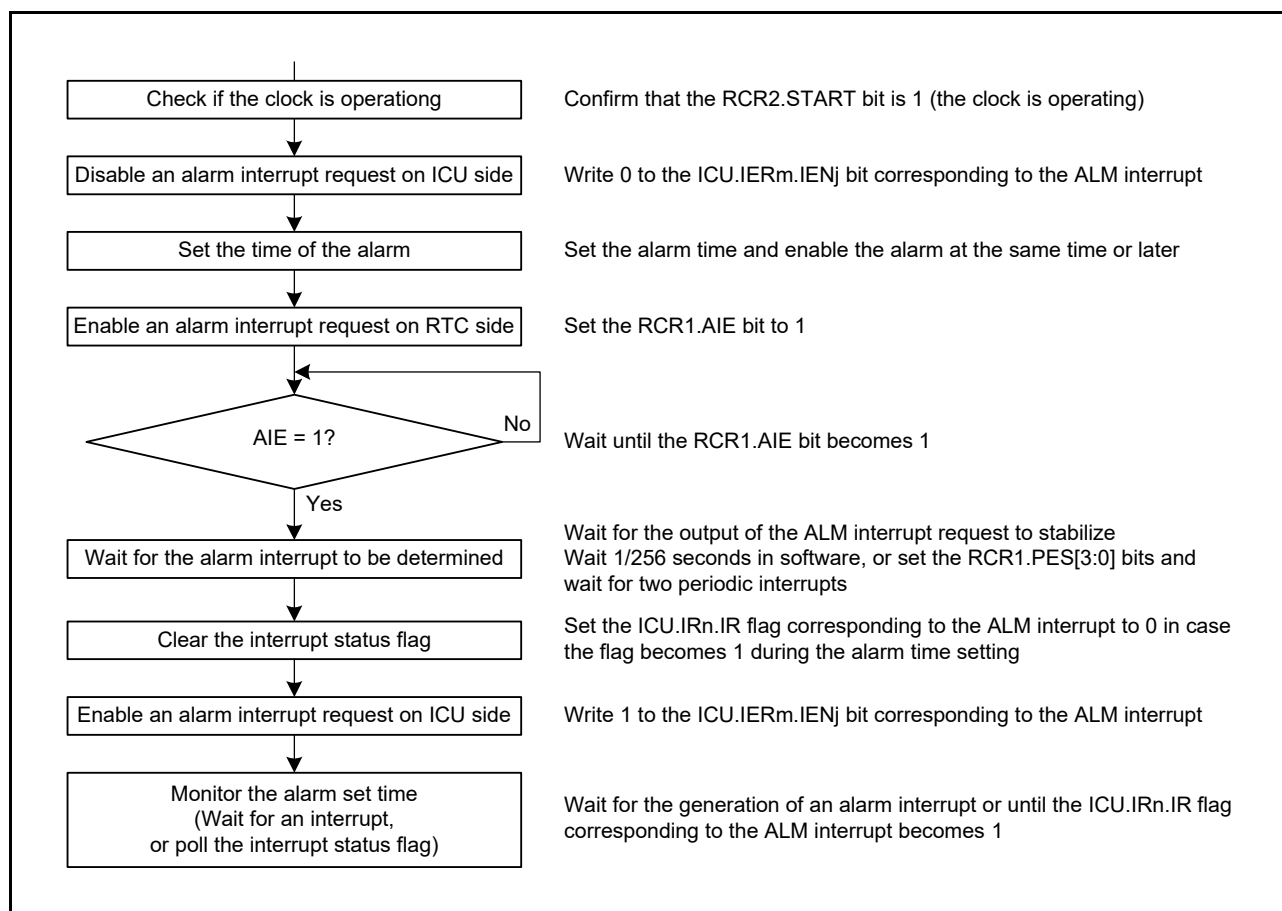


Figure 33.7 Using Alarm Function

In calendar count mode, an alarm can be generated by any one of year, month, date, day-of-week, hour, minute or second, or any combination of those. Write 1 to the ENB bit in the alarm registers involved in the alarm setting, and set the alarm time in the lower bits. Write 0 to the ENB bit in registers not involved in the alarm setting.

In binary count mode, an alarm can be generated in any bit combination of 32 bits. Write 1 to the ENB bit of the alarm enable register corresponding to the target bit of the alarm, and set the alarm time to the alarm register. For bits that are not target of the alarm, write 0 to the ENB bit of the alarm enable register.

When the counter and the alarm time match, the IR flag corresponding to the ALM interrupt is set to 1. Alarm detection can be confirmed by reading this bit, but an interrupt should be used in most cases. If 1 has been set in the interrupt request enable bit corresponding to the ALM interrupt, an alarm interrupt is generated in the event of alarm, enabling the alarm to be detected.

Writing 0 sets the IR flag corresponding to the ALM interrupt to 0.

When the counter and the alarm time match in a low power consumption state, the MCU returns from the low power consumption state. In deep software standby mode, the MCU returns from the deep software standby mode even when the alarm interrupt request is disabled.

33.3.7 Procedure for Disabling Alarm Interrupt

Figure 33.8 shows the procedure for disabling the enabled alarm interrupt request.

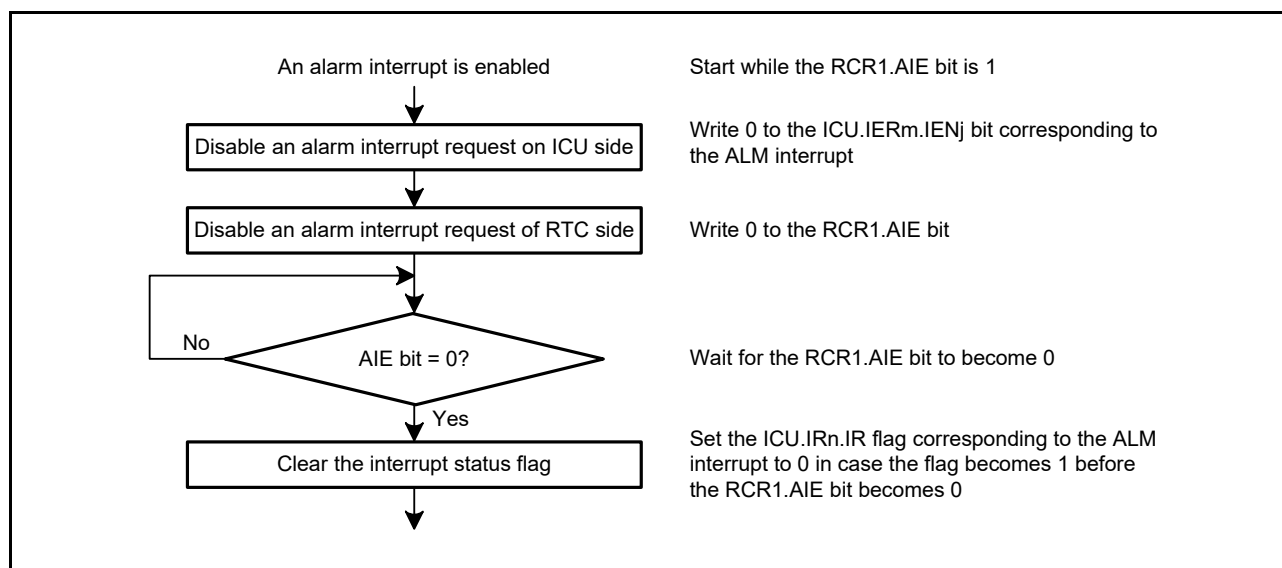


Figure 33.8 Procedure for Disabling Alarm Interrupt Request

33.3.8 Time Error Adjustment Function

The time error adjustment function is used to correct errors (running fast or slow) in the time due to the precision of oscillation by the sub-clock. Since 32,768 cycles of the sub-clock constitute 1 second of operation when the sub-clock is selected, the clock runs fast if the sub-clock frequency is high and slow if the sub-clock frequency is low. This function can be used to correct errors due to the clock running fast or slow.

Two types of time error adjustment functions are provided: automatic adjustment and adjustment by software. Use the RCR2.AADJE bit to select automatic adjustment or adjustment by software.

33.3.8.1 Automatic Adjustment

Enable automatic adjustment by setting the RCR2.AADJE bit to 1.

Automatic adjustment is the addition or subtraction of the value counted by the prescaler to or from the value in the RADJ register every time the adjustment period selected by the RCR2.AADJE bit elapses.

Examples are shown below.

[Example 1] Sub-clock running at 32.769 kHz

Adjustment procedure:

When the sub-clock is running at 32.769 kHz, 1 second elapses every 32,769 clock cycles. The RTC is meant to run at 32,768 clock cycles, so the clock runs fast by one clock cycle every second. The time on the clock is fast by 60 clock cycles per minute, so adjustment can take the form of setting the clock back by 60 cycles every minute.

Register settings: (when RCR2.CNTMD = 0)

- RCR2.AADJP = 0 (adjustment every minute)
- RADJ.PMADJ[1:0] = 10b (adjustment is performed by the subtraction from the prescaler.)
- RADJ.ADJ[5:0] = 60 (3Ch)

[Example 2] Sub-clock running at 32.766 kHz

Adjustment procedure:

When the sub-clock is running at 32.766 kHz, 1 second elapses every 32,766 clock cycles. The RTC is meant to run at 32,768 clock cycles, so the clock runs slow by two clock cycles every second. The time on the clock is slow by 20 clock cycles every 10 seconds, so adjustment can take the form of setting the clock forward by 20 cycles every 10 seconds.

Register settings: (when RCR2.CNTMD = 0)

- RCR2.AADJP = 1 (adjustment every 10 seconds)
- RADJ.PMADJ[1:0] = 01b (adjustment is performed by the addition to the prescaler.)
- RADJ.ADJ[5:0] = 20 (14h)

[Example 3] Sub-clock running at 32.764 kHz

Adjustment procedure:

At 32.764 kHz, 1 second elapses on 32,764 clock cycles. Since the RTC operates for 32,768 clock cycles as 1 second, the clock is delayed for four clock cycles per second. In 8 seconds, the delay is 32 clock cycles, so correction can be made by proceeding the clock for 32 clock cycles every 8 seconds.

Register settings when the RCR2.CNTMD bit is 1

- RCR2.AADJP = 1 (adjustment every 8 seconds)
- RADJ.PMADJ[1:0] = 01b (adjustment is performed by the addition to the prescaler.)
- RADJ.ADJ[5:0] = 32 (20h)

33.3.8.2 Adjustment by Software

Enable adjustment by software by setting the RCR2.AADJE bit to 0.

Adjustment by software is the addition or subtraction of the value counted by the prescaler to or from the value in the RADJ register at the time of execution of an instruction for writing to the RADJ register.

An example is shown below.

[Example 1] Sub-clock running at 32.769 kHz

Adjustment procedure:

When the sub-clock is running at 32.769 kHz, 1 second elapses every 32,769 clock cycles. The RTC is meant to run at 32,768 clock cycles, so the clock runs fast by one clock cycle every second. The time on the clock is fast by one clock cycle per second, so adjustment can take the form of setting the clock back by one cycle every second.

Register settings:

- RADJ.PMADJ[1:0] = 10b (adjustment is performed by the subtraction from the prescaler.)
- RADJ.ADJ[5:0] = 1 (01h)

This is written to the RADJ register once per 1-second interrupt.

33.3.8.3 Procedure for Changing the Mode of Adjustment

When changing the mode of adjustment, change the value of the AADJE bit in RCR2 after setting the RADJ.PMADJ[1:0] bits to 00b (adjustment is not performed).

Changing from adjustment by software to automatic adjustment:

- (1) Set the RADJ.PMADJ[1:0] bits to 00b (adjustment is not performed).
- (2) Set the RCR2.AADJE bit to 1 (automatic adjustment is enabled).
- (3) Use the RCR2.AADJP bit to select the period of adjustment.
- (4) In RADJ, set the PMADJ[1:0] bits for addition or subtraction and the ADJ[5:0] bits to the value for use in time error adjustment.

Changing from adjustment by software to automatic adjustment:

- (1) Set the RADJ.PMADJ[1:0] bits to 00b (adjustment is not performed).
- (2) Set the RCR2.AADJE bit to 0 (adjustment by software is enabled).
- (3) Proceed with adjustment by setting the RADJ.PMADJ[1:0] bits for addition or subtraction and the RADJ.ADJ[5:0] bits to the value for use in time error adjustment at the desired time. After that, the time is adjusted every time a value is written to the RADJ register.

33.3.8.4 Procedure for Stopping Adjustment

Stop adjustment by setting the RADJ.PMADJ[1:0] bits to 00b (adjustment is not performed).

33.3.9 Time Capture Function

The RTC is capable of storing the month, date, hour, minute and second/binary counters 3 to 0 by detecting an edge of a signal on a time capture event input pin.

A noise filter can also be used on a time capture event input pin. If the noise filter is enabled, the TCST flag is set to 1 when the input level on the pin matches three times.

The noise filter can be switched on or off for each of the time capture event input pins. Operation when the noise filter is off is shown in Figure 33.9 and operation when the noise filter is on is shown in Figure 33.10.

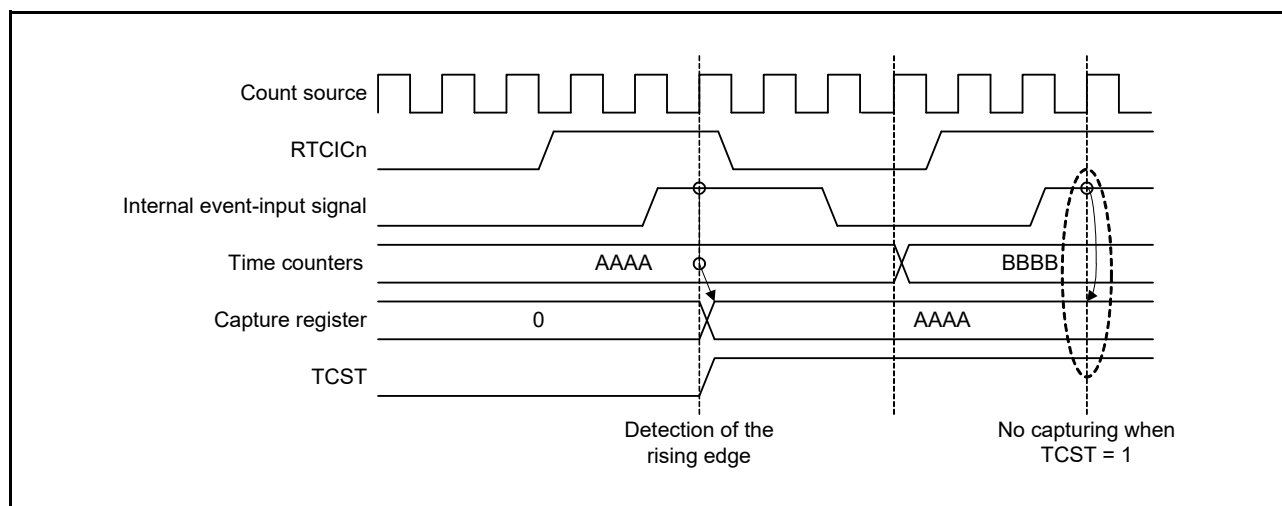


Figure 33.9 Timing of a Time Capture Operation (with the Filter Off) (n = 0 to 2)

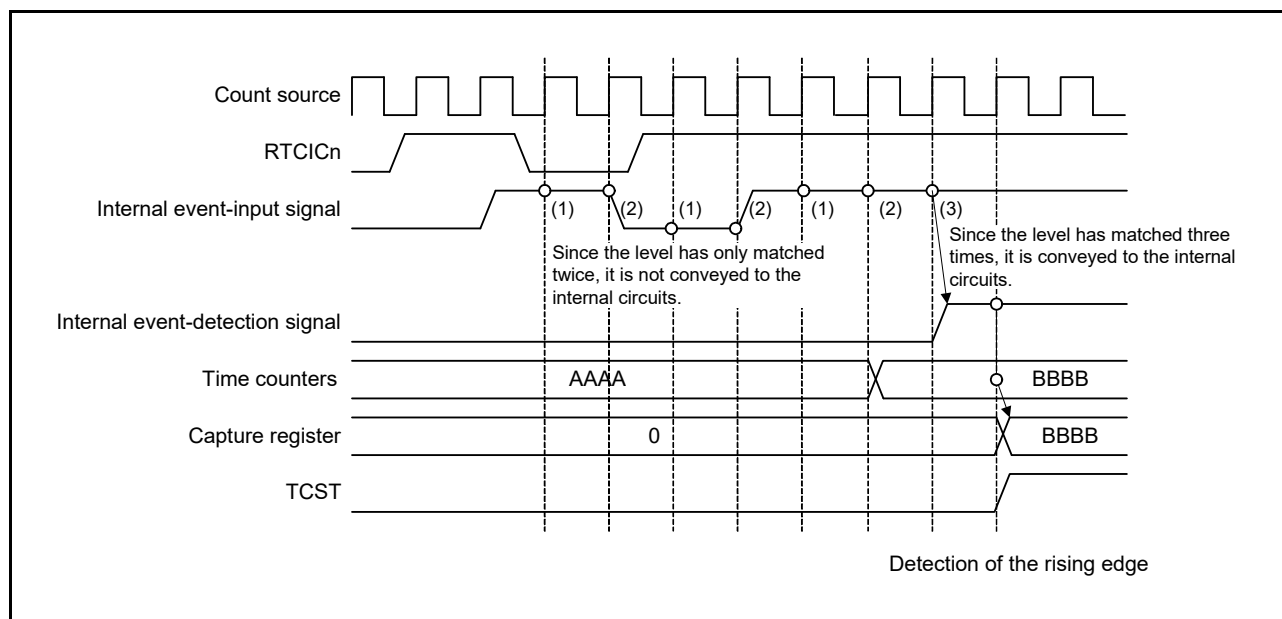


Figure 33.10 Timing of a Time Capture Operation (with the Filter On) (n = 0 to 2)

33.4 Interrupt Sources

There are three interrupt sources in the realtime clock. Table 33.4 lists interrupt sources for the RTC.

Table 33.4 RTC Interrupt Sources

Name	Interrupt Sources
ALM	Alarm interrupt
PRD	Periodic interrupt
CUP	Carry interrupt

(1) Alarm interrupt (ALM)

This interrupt is generated according to the result of comparison between the alarm registers and realtime clock counters (for details, refer to section 33.3.6, Alarm Function).

Since there is a possibility that the interrupt flag may be set to 1 when the settings of the alarm registers match the clock counters, wait for the alarm time settings to be confirmed and set the IR flag corresponding to the ALM interrupt to 0 again after modifying values of the alarm registers. Once the interrupt flag for the alarm interrupt has been set to 1 and the state has returned to non-matching of the alarm registers and clock counters, the flag will not be set again until there is a further match or the values of the alarm registers are modified again.

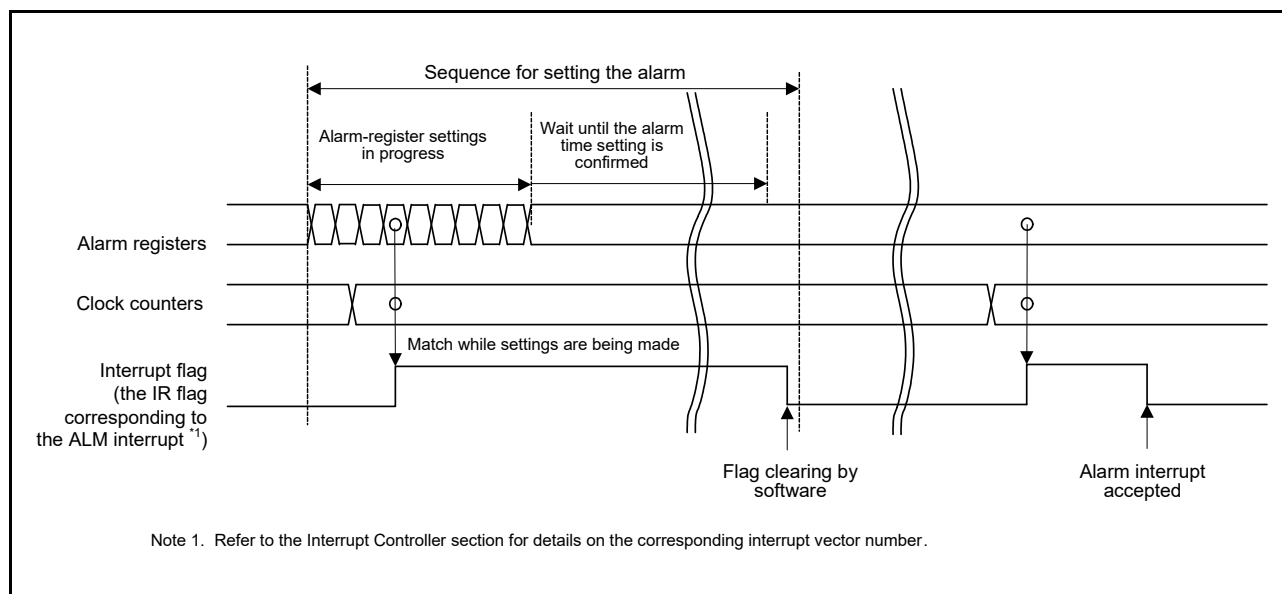


Figure 33.11 Timing Chart for the Alarm Interrupt (ALM)

(2) Periodic interrupt (PRD)

This interrupt is generated at intervals of 2 seconds, 1 second, 1/2 second, 1/4 second, 1/8 second, 1/16 second, 1/32 second, 1/64 second, 1/128 second, or 1/256 second. The interrupt interval can be selected through the RCR1.PES[3:0] bits.

(3) Carry interrupt (CUP)

This interrupt is generated when a carry to the second counter/binary counter 0 occurred or a carry to the R64CNT counter occurred during read access to the 64-Hz counter.

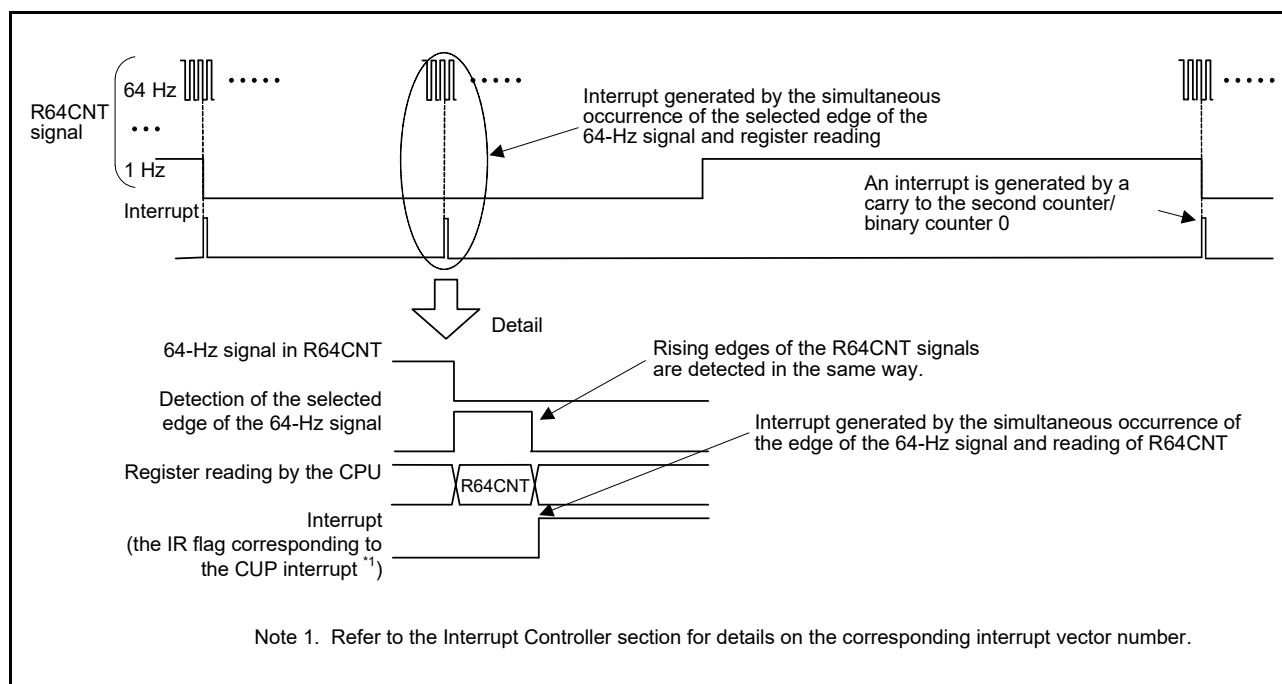


Figure 33.12 Carry Interrupt (CUP) Timing Chart

33.5 Event Link Output

The RTC outputs the following event signals for the event link controller (ELC), and these can be used to initiate operations by other modules selected in advance.

(1) Periodic event output

The periodic event signal is output at the interval selected from among 1/256, 1/128, 1/64, 1/32, 1/16, 1/8, 1/4, 1/2, 1, and 2 seconds by the setting of the RCR1.PES[3:0] bits.

The event generation period immediately after the event generation is selected is not guaranteed.

Note: If event linking from the RTC is to be used, only make the ELC settings after making the RTC settings (initialization, time settings, etc.). Making the RTC settings after the ELC settings can lead to the output of unexpected event signals.

33.5.1 Interrupt Handling and Event Linking

The RTC has a bit to enable or disable periodic interrupts. An interrupt request signal is output for the CPU when an interrupt source is generated while the corresponding enable bit is enabled.

In contrast, an event link output signal is sent to other modules as an event signal via the ELC when an interrupt source is generated, regardless of the setting of the corresponding interrupt enable bit.

Note: Although alarm and periodic interrupts can still be output during software standby or deep software standby, the periodic event signals for the ELC are not output.

33.6 Usage Notes

33.6.1 Register Writing during Counting

The following registers should not be written to during counting (while the RCR2.START bit = 1).

RSECCNT/BCNT0, RMINCNT/BCNT1, RHRCNT/BCNT2, RDAYCNT, RWKCNT/BCNT3, RMONCNT, RYRCNT, RCR1.RTCOS, RCR2.RTCOE, RCR2.HR24, RFRH, RFRL

The counter must be stopped before writing to any of the above registers.

33.6.2 Use of Periodic Interrupts

The procedure for using periodic interrupts is shown in Figure 33.13.

The generation and period of the periodic interrupt can be changed by the setting of the RCR1.PES[3:0] bits. However, since the prescaler, R64CNT, and RSECCNT/BCNT0 are used to generate interrupts, the interrupt period is not guaranteed immediately after setting of the RCR1.PES[3:0] bits.

Furthermore, stopping/restarting or resetting counter operation, reset by RTC software, and the 30-second adjustment by changing the RCR2 value affects the interrupt period. When the time error adjustment function is used, the interrupt generation period after adjustment is added or subtracted according to the adjustment value.

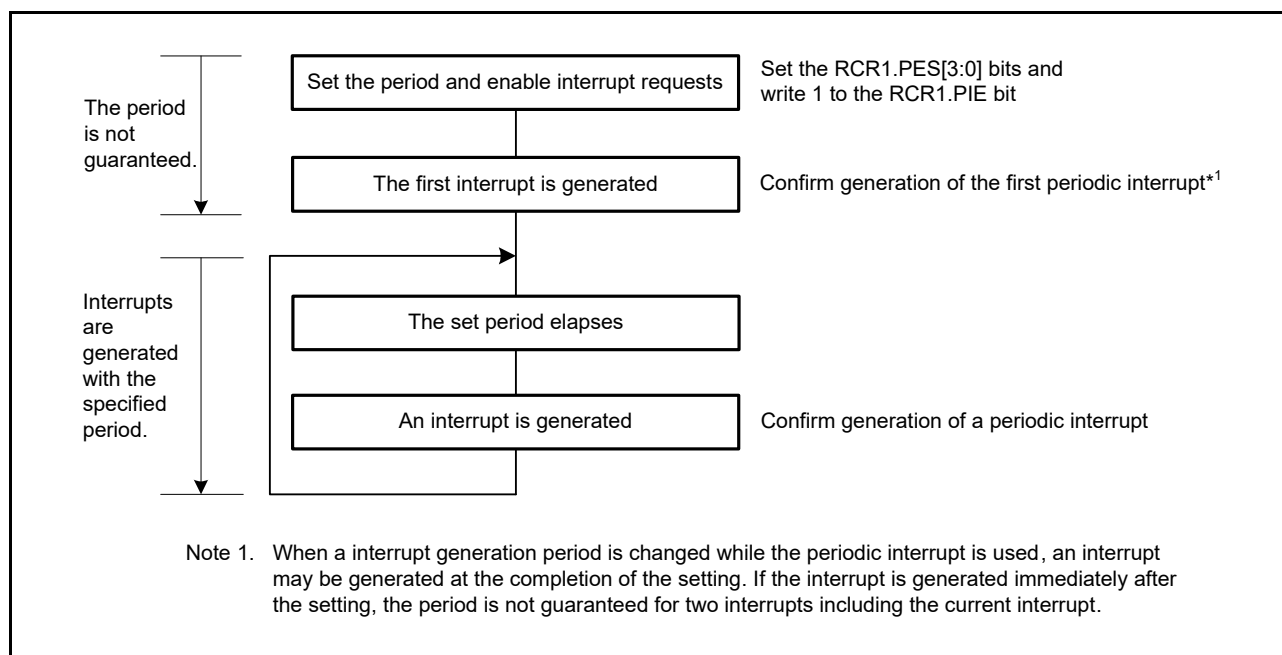


Figure 33.13 Using Periodic Interrupt Function

33.6.3 RTCOUT (1-Hz/64-Hz) Clock Output

Stopping/restarting or resetting counter operation, reset by RTC software, and the 30-second adjustment by changing the RCR2 value affects the period of RTCOUT (1-Hz/64-Hz) output. When the time error adjustment function is used, the period of RTCOUT (1-Hz/64-Hz) output after adjustment is added or subtracted according to the adjustment value.

33.6.4 Transitions to Low Power Consumption Modes after Setting Registers

A transition to a low power consumption state (software standby mode, deep software standby mode, or battery backup) during writing to or updating of an RTC register might destroy the register's value. After setting a register, confirm that the setting is in place before initiating a transition to a low power consumption state.

33.6.5 Notes on Writing to and Reading from Registers

- When reading a counter register such as the second counter/binary counter 0 after having written to the counter register, follow the procedure in [section 33.3.5, Reading 64-Hz Counter and Time](#).
- The value written to the count registers, alarm registers, year alarm enable register, bits RCR2.AADJE, AADJP, and HR24, RCR3, RCR4, RFRH, or RFRL register is reflected when four read operations are performed after writing.
- The values written to the RCR1.CIE, RTCOS, and RCR2.RTCOE bits can be read immediately after writing.
- To read the value from the timer counter after return from a reset, deep software standby mode, software standby mode, or the battery backup state, wait for 1/128 second while the clock is operating (RCR2.START bit = 1).
- After a reset is generated, write to the RTC register when six cycles of the count source have elapsed.

33.6.6 Changing the Count Mode

When changing the count mode (calendar/binary), set the RCR2.START bit to 0, stop counting operation, then start again from the initial setting. For details on initial setting, refer to [section 33.3.1, Outline of Initial Settings of Registers after Power On](#).

33.6.7 Initialization Procedure When the Realtime Clock is Not to be Used

Registers in the RTC are not initialized by a reset. Accordingly, depending on the initial state, the generation of an unintentional interrupt request or operation of the counter may lead to increased power consumption.

For products that do not require a realtime clock, initialize the registers by following the initialization procedure shown in Figure 33.14.

Alternatively, when the sub-clock is not used as the system clock or realtime clock, the counter can be stopped by writing 0 (sub-clock oscillator is selected) to the RCR4.RCKSEL bit and stopping the sub-clock.

When making the setting to stop the sub-clock, write 0 to the RCR3.RTCEN bit and write 1 to the SOSCCR.SOSTP bit. For details on the setting of the SOSCCR.SOSTP bit, refer to section 9, Clock Generation Circuit.

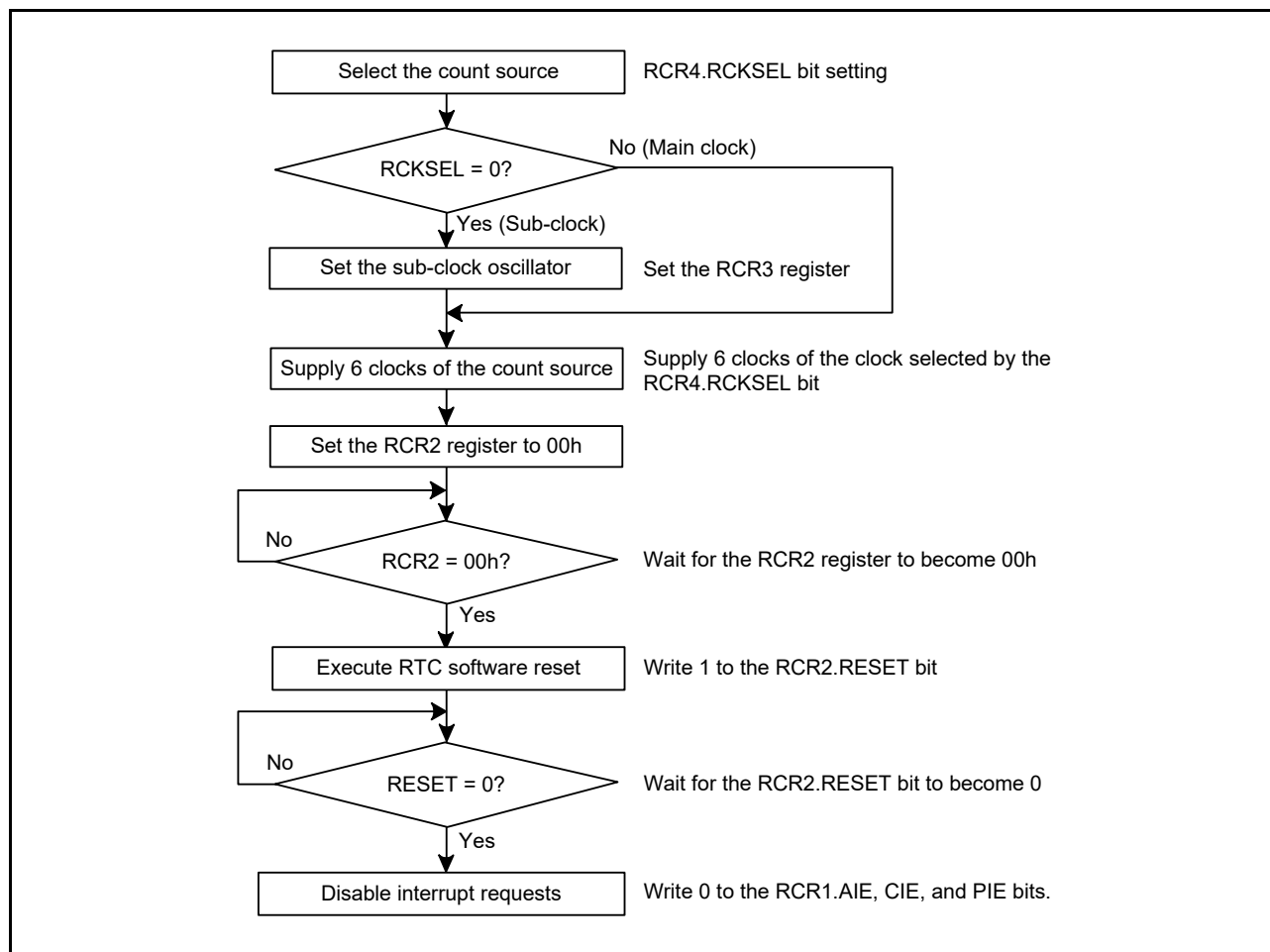


Figure 33.14 Initialization Procedure